Anytime-valid detection of volatility changes in high-frequency returns with robust truncation

Shreyan Thanki
Draft, November 2026

Abstract

We study the online detection of changes in the volatility of one-minute returns when the returns contain jumps and fat tails and the monitor is inspected after every observation. Following Bapat (2026), each return is clipped at c times a robust scale, the median absolute deviation (MAD), which we estimate from past data only. We monitor the clipped squared return with Shiryaev-Roberts betting e-detectors (Shin, Ramdas and Rinaldo 2024), one for increases and one for decreases. With a known baseline, the average run length to a false alarm is at least A and the probability of any false alarm within m steps is at most m/A, for every return distribution and every form of dependence that keeps the conditional mean of the clipped squared return on the null side. A one-sided form of the impossibility result of Bahadur and Savage (1956) shows that no nontrivial e-process exists for the null "the variance has not fallen"; a bounded statistic such as the clipped squared return is needed to detect decreases, and the clipping level then sets a speed limit for them. With a baseline estimated by a confidence sequence the guarantee becomes $\delta + m/A$. We give a non-asymptotic bound on the detection delay, a decomposition of the delay cost of robustness into a truncation loss, a nonparametric loss and a grid loss, and a closed-form cap on the signal that remains after dividing returns by a GARCH forecast. In simulations with the true baseline the bound holds under jumps, Student t returns and price ticks (1.1% to 1.4% false alarms per day against a bound of 5%), while Gaussian likelihood-ratio, CUSUM and Bayesian online change-point detectors raise false alarms on half or more of the days under jumps or fat tails; on clean Gaussian data the Bayesian detector is faster, especially for large decreases. On five cryptocurrencies the alarms carry information beyond EWMA and HAR volatility forecasts, and on the Nifty 50 index and around FOMC statements they follow scheduled events far more often than control times do. A ratio rule tuned with hindsight is about as precise, and published change-point methods given day-long windows come close: on real data the contribution is settings fixed in advance and, with the confidence-sequence baseline, a proved false alarm bound, not extra precision.

Keywords. Sequential change detection; e-processes; testing by betting; high-frequency volatility; truncation; jumps; anytime validity.

Status of this draft. Every number below comes from an experiment of the accompanying repository, cited by its label (for example "(E9)"). Each result is marked **[SYNTHETIC]** (our simulation), **[REAL]** (our run on market data) or **[PUBLISHED]** (printed in Bapat 2026).

1 Introduction

Volatility in high-frequency markets moves between regimes: calm, stressed, calm again. Risk limits, position sizes and model refits should react when it moves. This paper asks for

an online monitor that reads a stream of one-minute returns and raises an alarm when volatility shifts up or down, with a proven limit on false alarms that holds however often the monitor is checked. Two obstacles make this hard.

Jumps and fat tails. A few extreme returns dominate sums of squared returns, and even more so sums of fourth powers. A detector built on a Gaussian likelihood treats a single jump as strong evidence of higher volatility. Jump-robust high-frequency estimators address this for a fixed sample: bipower variation (Barndorff-Nielsen and Shephard 2004), nearest-neighbour truncation (Andersen, Dobrev and Schaumburg 2012), threshold estimators (Mancini 2009), and the kernel quarticity estimator of Bapat (2026), which clips every return at c times its MAD before smoothing, so that no single return has unbounded influence.

Repeated looks. A fixed-sample test applied at every minute will eventually reject by chance. In our simulations a 60-step z-test repeated at every step raised a false alarm within one 390-step day on 71% of Gaussian paths (E4) [SYNTHETIC]. Classical sequential change detection (Page 1954; Shiryaev 1963; Roberts 1966; Lorden 1971; Pollak 1985) controls the average run length to a false alarm, but it needs the pre-change law, usually in parametric form. E-detectors (Shin, Ramdas and Rinaldo 2024) give the same control for composite, nonparametric nulls, using e-processes built by betting (Shafer 2021; Ramdas, Grünwald, Vovk and Shafer 2023; Waudby-Smith and Ramdas 2024).

The robust estimators solve the first problem for a fixed sample and do not address the second. Anytime-valid methods solve the second, but they need a statistic whose conditional mean has a meaning under the null and, for the direction “volatility went down”, a bounded one. Truncation, used for jump robustness since Mancini (2009) and in the MAD-based form of Bapat (2026), supplies exactly such a statistic. This paper puts the two together.

What is known and what is ours. The building blocks are known: Ville’s (1939) inequality, test supermartingales and testing by betting (Shafer 2021; Waudby-Smith and Ramdas 2024), e-detectors and their run-length guarantee (Shin, Ramdas and Rinaldo 2024), empirical Bernstein confidence sequences (Howard, Ramdas, McAuliffe and Sekhon 2021; Waudby-Smith and Ramdas 2024), the impossibility result of Bahadur and Savage (1956), and Lorden’s (1971) and Lai’s (1998) delay lower bounds. Our contributions are the following.

- **A robust statistic and null for online volatility monitoring.** We monitor $X_t = \min(r_t^2, \tau^2)$, the clipped squared return, with a truncation level $\tau = c\,\hat{s}$ computed from past data only ($\hat{s}$ is the scaled MAD). The nulls restrict only the conditional mean of X_t. Under a fixed innovation shape this mean is a nondecreasing function of the conditional volatility (Lemma 1), so the nulls have a volatility meaning without a model for the return distribution.
- **Validity, with exact conditions.** For Shiryaev-Roberts betting e-detectors on X_t we state and prove the run-length bound $\mathbb{E}[T_A] \geq A$ (Theorem A, an instance of Shin, Ramdas and Rinaldo 2024), the finite-horizon bound $\mathbb{P}(T_A \leq m) \leq m/A$ (Theorem B), a two-sided rule, and a bound on the expected number of false alarms under recalibration after every alarm (Corollary C).
- **Why a bounded statistic.** A one-sided form of the Bahadur and Savage (1956) impossibility (Proposition 2, Corollary 3): every e-process for the sequential null

"the conditional variance has not fallen" is bounded by 1, so no e-detector can detect a fall in the variance itself. A bounded functional is needed; the truncated variance is one with a direct volatility meaning, and truncation then imposes an explicit speed limit on decrease detection (Corollary 7). The argument is elementary and we do not claim it is new; its role in this application is the point.

- **An estimated baseline.** We measure how much a plug-in baseline inflates the false alarm rate, and we prove $\mathbb{P}(\text{false alarm within } m) \le \delta + m/A$ for a baseline estimated by a confidence sequence (Theorem D), with a version for recalibration after every alarm. The confidence sequence is the predictable plug-in empirical Bernstein construction of Waudby-Smith and Ramdas (2024) with one-sided ranges; its use as the baseline of a detector, with this bound, is what we add.
- **Detection delay.** A non-asymptotic bound on the worst-case delay (Theorem 8); first-order asymptotic optimality of a refined grid of bets among all rules that control the run length over the nonparametric null, a known result for bounded means (Ram and Ramdas 2026) for which we give the constants of this detector (Theorem 9); and a decomposition of the delay cost of robustness into a truncation loss, a nonparametric loss and a grid loss (Corollary 6). Under GARCH clustering, monitoring returns divided by a correct volatility forecast makes the null hold again, but caps the visible signal of a variance break at $(1-\beta)/\alpha$ (Proposition 10, closed form).
- **Evidence.** Simulations of validity and delay against Gaussian likelihood-ratio, CUSUM and repeated-test benchmarks and against two published change-point methods, and of the failure modes (clustering, estimated baselines). Real-data studies on the five cryptocurrencies of Bapat (2026), the Nifty 50 index with India VIX, and FOMC and CPI releases, with random-time, forecast and rate-matched controls. First results on a running truncation level and on restarts without a blind period, on combining detectors at 1, 5 and 15 minutes, and on a panel of assets with false discovery control.

We also report, briefly and as questions, what an independent reproduction of the published tables of Bapat (2026) found about the estimator (Section 7). We do not claim to have improved the estimator.

Related work. Retrospective tests for volatility breaks include the cumulative sums of squares of Inclán and Tiao (1994) and the high-frequency, nonparametric and minimax-optimal test of Bibinger, Jirak and Vetter (2017). Online monitoring for a switch to explosive GARCH volatility on daily data is studied by Horváth, Trapani and Wang (2024). Andersen, Todorov and Zhou (2025) monitor high-frequency prices in real time for local departures from the semimartingale model, a different target. Bayesian online change-point detection (Adams and MacKay 2007) is a standard online method that models the data inside each regime; we compare with it and with a test of the Bibinger, Jirak and Vetter type in Section 5.6. Our literature search (November 2026, not exhaustive) did not find a method that combines jump-robust high-frequency volatility measures with anytime-valid sequential change detection; we state this with that caveat.

Organisation. Section 2 sets up the statistic. Section 3 gives the detectors and their guarantees, the impossibility result and the estimated baseline. Section 4 studies the detection delay. Sections 5 and 6 report simulation and real-data evidence. Section 7 summarises the reproduction of the estimator. Section 8 discusses limitations and open

problems. Longer proofs and settings are in the appendices, and a supplement gives detailed versions of several results that the main text summarises.

2 Setting and the truncated statistic

Returns and information. Let r_t be the one-minute log return ending at minute t. Monitoring starts after a calibration window $r_{-n+1}, \ldots, r_0$; we write $\mathcal{F}_t = \sigma(r_{-n+1}, \ldots, r_t)$ for the information up to time t. A quantity is *predictable* if its value used at time t is $\mathcal{F}_{t-1}$-measurable. Everything the monitor uses (truncation level, baseline, bets, intraday profile) must be predictable.

The truncation of Bapat (2026). Bapat (2026, eq. 11) clips returns at $\tau = c\,\mathrm{MAD}(r)$ with $c = 3$, $\tilde{r}_i = \max(-\tau, \min(r_i, \tau))$, where the MAD is computed over the whole sample, and then forms the kernel-weighted local quarticity $\hat{Q}_t = \frac{1}{3}\sum_{i=t-h}^{t} K((i-t)/h)\,\tilde{r}_i^4$ (eq. 12) and its average over t (eq. 16). We keep the clipping and change three things.

1. *Scaled MAD.* We use $\hat{s} = 1.4826\,\mathrm{median}_i|r_i|$, the MAD about zero scaled to estimate a Gaussian standard deviation (the median of one-minute returns is 0 in all our data sets, E6, E7 [REAL]). With $\tau = c\,\hat{s}$ and Gaussian returns the cut-off sits at c standard deviations. With the raw MAD it sits at $0.6745\,c$ standard deviations.
2. *Past data only.* $\hat{s}$ and τ are computed from the calibration window, so τ is $\mathcal{F}_0$-measurable and predictable for every monitored t. A rolling or exponentially weighted MAD keeps the jump protection of the full-sample MAD in simulation (E3) [SYNTHETIC]: the 95th percentile of the absolute log error of a local variance estimate falls from about 1.4 without truncation to about 0.52 with an online MAD.
3. *The squared, not the fourth-power, statistic.* We monitor

   $$X_t = \min(r_t^2, \tau^2) = \tilde{r}_t^2 \in [0, \tau^2],$$

   the summand of truncated realised variance, which is the second-moment analogue of eq. (12) with bandwidth $h = 0$. The same detectors applied to $\min(r_t^4, \tau^4)$ are valid as well; for pure variance changes they were 1% to 28% faster for increases and 1.8 to 2.6 times slower for decreases (ALT1) [SYNTHETIC]. The local truncated quarticity enters the variance monitor only through the variance of X_t, which governs bet sizing (Section 4).

Truncated moments. For $Z \sim N(0,1)$ and a cut-off $a > 0$, write φ, Φ for its density and distribution function and

$$\kappa_2(a) = \mathbb{E}[\min(Z^2, a^2)] = 2\Phi(a) - 1 - 2a\varphi(a) + 2a^2(1 - \Phi(a)),$$
$$\kappa_4(a) = \tfrac{1}{3}\mathbb{E}[\min(Z^4, a^4)] = \tfrac{1}{3}[3(2\Phi(a) - 1 - 2a\varphi(a)) - 2a^3\varphi(a) + 2a^4(1 - \Phi(a))].$$

For $c = 3$: with the scaled MAD ($a = 3$), 0.27% of Gaussian returns are clipped, $\kappa_2 = 0.995$ and $\kappa_4 = 0.964$; with the raw MAD ($a = 2.023$), 4.3% are clipped, $\kappa_2 = 0.925$ and $\kappa_4 = 0.705$ (E2) [SYNTHETIC agreement with the closed form]. For Gaussian returns with standard deviation σ_0 and the scaled MAD, the null mean of X_t is $m_0 = \kappa_2(c)\sigma_0^2$ and the ratio $B = \tau^2/m_0 = c^2/\kappa_2(c)$ equals 9.045 for $c = 3$. The constant B, the largest possible value of X_t/m_0, governs the decrease detector.

Intraday seasonality and missing returns. Intraday volatility follows a strong pattern (Andersen and Bollerslev 1997). Before monitoring, each return is divided by a predictable profile: for crypto, the median of $|r|$ at the same minute of the day over the

previous 20 days, smoothed across neighbouring minutes; for the Nifty 50, a session-aware version that does not wrap the close into the open (E7). Medians limit the effect of jumps on the profile (Boudt, Croux and Laurent 2011). The profile is treated as known; errors in it act as a small violation of the null that no experiment isolates (Section 8). A missing return is skipped: the detection statistic is left unchanged at that step. Validity is kept when, given the past, whether a return is missing says nothing about its size; Lemma 2 below then holds in the form $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] \le M_{t-1} + \mathbb{P}(r_t \text{ observed} \mid \mathcal{F}_{t-1}) \le M_{t-1} + 1$, and the horizons m in Theorems B and D count every monitored step, with or without a return. Missing returns must not be replaced by zeros, which look like evidence for lower volatility.

3 Betting e-detectors on the truncated statistic

3.1 Nulls and their volatility meaning

Let $\mu_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ be the conditional truncated variance, and let $m_0 > 0$ be a baseline. For a band $b \in [0, 1)$ fixed in advance, the nulls are

$$H_0^{\uparrow}\colon \mu_t \le (1+b)\, m_0 \quad \text{for all } t \ge 1, \qquad H_0^{\downarrow}\colon \mu_t \ge (1-b)\, m_0 \quad \text{for all } t \ge 1.$$

They restrict one conditional moment and allow any return distribution, any jump process and any dependence that keeps that moment on the null side. “Any dependence” is a statement about the shape of the joint law, not a licence for any data: the conditional mean must stay on the null side at every step. Persistent volatility clustering and price-level microstructure noise both move it, even when the unconditional variance is constant (Sections 5.4 and 8).

Lemma 1 (volatility meaning). Suppose $r_t = \sigma_t \varepsilon_t$, where $\sigma_t > 0$ is $\mathcal{F}_{t-1}$-measurable and ε_t is independent of $\mathcal{F}_{t-1}$ with a law G that does not depend on t. Let τ be $\mathcal{F}_0$-measurable and $g_\tau(s) = \mathbb{E}_G[\min(s^2\varepsilon^2, \tau^2)]$. Then $\mu_t = g_\tau(\sigma_t)$ and g_τ is nondecreasing on $(0, \infty)$. Hence, with $m_0 = g_\tau(\sigma_0)$ and $b = 0$, $H_0^{\uparrow}$ holds whenever $\sigma_t \le \sigma_0$ for all t, and $H_0^{\downarrow}$ whenever $\sigma_t \ge \sigma_0$ for all t.

Proof. For each fixed ε, $s \mapsto \min(s^2\varepsilon^2, \tau^2)$ is nondecreasing. Since σ_t and τ are $\mathcal{F}_{t-1}$-measurable and ε_t is independent of $\mathcal{F}_{t-1}$, $\mu_t = g_\tau(\sigma_t)$, and monotonicity passes through the expectation. ▫

Lemma 1 holds for every innovation law G: only the size of the change in μ_t for a given change in σ_t depends on G. Two caveats follow from it. A change in the tail shape alone, with σ_t fixed, also moves μ_t and can trigger an alarm (with $c = 3$, a switch from Gaussian to Student t innovations at the same variance lowers μ_t and is seen as a volatility decrease, ALT1 [SYNTHETIC]). And without the independence of ε_t and σ_t (for example under a contemporaneous leverage effect) μ_t need not be monotone in σ_t.

3.2 Bets, Shiryaev-Roberts e-detectors and the false alarm guarantees

Write $Y_t = X_t / m^{\uparrow}$ with $m^{\uparrow} = (1+b)m_0$ for the increase side, and $Y_t = X_t / m^{\downarrow}$ with $m^{\downarrow} = (1-b)m_0$ and $B = \tau^2 / m^{\downarrow} > 1$ for the decrease side, so that $Y_t \in [0, B]$. With a predictable bet λ_t define the betting factors

$$f_t^{\uparrow} = 1 + \lambda_t (Y_t - 1), \ \ \lambda_t \in [0, 1), \qquad f_t^{\downarrow} = 1 + \lambda_t (1 - Y_t), \ \ \lambda_t \in [0, 1/(B-1)).$$

Since $Y_t \ge 0$, $f_t^{\uparrow} \ge 1 - \lambda_t > 0$, and since $Y_t \le B$, $f_t^{\downarrow} \ge 1 - \lambda_t(B-1) > 0$. Under the corresponding null, $\mathbb{E}[f_t \mid \mathcal{F}_{t-1}] = 1 + \lambda_t(\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] - 1) \le 1$ for the increase side, and symmetrically for the decrease side. Hence for every start time j the wealth $L_t^{(j)} = \prod_{s=j}^{t} f_s$ (with $L_{j-1}^{(j)} = 1$) is a nonnegative supermartingale with initial value 1, a *test supermartingale*.

Definition (Shiryaev-Roberts e-detector). Fix bets $\lambda_1, \dots, \lambda_K$ in the admissible range and weights $w_k > 0$ with $\sum_k w_k = 1$. For each k let $M_t^{(k)} = f_t^{(k)}(M_{t-1}^{(k)} + 1)$ with $M_0^{(k)} = 0$, so that $M_t^{(k)} = \sum_{j=1}^{t} \prod_{s=j}^{t} f_s^{(k)}$ is the total wealth of bettors who start at every time $j \le t$. Let $M_t = \sum_k w_k M_t^{(k)}$ and raise an alarm at $T_A = \inf\{t \ge 1 : M_t \ge A\}$.

The default grids are $\lambda \in \{0.02, 0.05, 0.1, 0.2, 0.35, 0.5\}$ for increases and $\lambda/\lambda_{\max} \in \{0.1, 0.25, 0.5, 0.75, 0.95\}$ with $\lambda_{\max} = 1/(B-1)$ for decreases, with equal weights. The updates cost $O(K)$ per step (about 10 microseconds per step for the whole streaming monitor in plain Python and NumPy, E9) [SYNTHETIC].

Lemma 2. Under the null, $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] \le M_{t-1} + 1$ for every $t \ge 1$, so $M_t - t$ is a supermartingale. The same holds for any predictable bets, and for any average of such statistics with fixed weights.

Proof. $\mathbb{E}[M_t^{(k)} \mid \mathcal{F}_{t-1}] = \mathbb{E}[f_t^{(k)} \mid \mathcal{F}_{t-1}](M_{t-1}^{(k)} + 1) \le M_{t-1}^{(k)} + 1$; average over k. ▫

Theorem A (average run length). Under the null, $\mathbb{E}[T_A] \ge A$ for every $A > 0$.

Proof. If $\mathbb{E}[T_A] = \infty$ there is nothing to show; otherwise $T_A < \infty$ almost surely. For fixed t, optional stopping of the supermartingale $M_t - t$ (Lemma 2) at the bounded time $T_A \wedge t$ gives $\mathbb{E}[M_{T_A \wedge t}] \le \mathbb{E}[T_A \wedge t] \le \mathbb{E}[T_A]$. As $t \to \infty$, $M_{T_A \wedge t} \to M_{T_A} \ge A$ almost surely, and since $M \ge 0$, Fatou's lemma gives $A \le \mathbb{E}[M_{T_A}] \le \operatorname{liminf}_t \mathbb{E}[M_{T_A \wedge t}] \le \mathbb{E}[T_A]$. ▫

Theorem A is the run-length guarantee of e-detectors (Shin, Ramdas and Rinaldo 2024), who define an e-detector by $\mathbb{E}[M_\rho] \le \mathbb{E}[\rho]$ for all stopping times ρ; we include the proof to make the conditions explicit. A practitioner usually needs a statement for a finite horizon, such as one trading day.

Theorem B (false alarm within a horizon). Under the null, $\mathbb{P}(T_A \le m) \le m/A$ for every integer $m \ge 1$.

Proof. For $0 \le t \le m$ let $S_t = M_t + (m - t)$. By Lemma 2, S is a supermartingale on $\{0, \dots, m\}$; it is nonnegative and $S_0 = m$. Ville's inequality gives $\mathbb{P}(\max_{t \le m} S_t \ge A) \le m/A$, and $M_t \ge A$ implies $S_t \ge A$. ▫

The argument is elementary and we do not claim it is new. Choosing $A = 7800$ for a 390-step trading day, or $A = 28{,}800$ for a 1440-minute crypto day, bounds the probability of any false alarm per day by 5%. The CUSUM-type statistic $C_t = f_t \max(C_{t-1}, 1)$ satisfies $C_t \le M_t$, so it alarms no earlier and inherits both guarantees.

Proposition 1 (two-sided monitoring). Under the two-sided null $(1-b)m_0 \le \mu_t \le (1+b)m_0$ for all t, the average $\bar{M}_t = (M_t^{\uparrow} + M_t^{\downarrow})/2$ satisfies Lemma 2, and $\max(M_t^{\uparrow}, M_t^{\downarrow}) \ge 2A$ implies $\bar{M}_t \ge A$. Hence the rule that alarms at the first t with $\max(M_t^{\uparrow}, M_t^{\downarrow}) \ge 2A$ satisfies Theorems A and B, and the side that crosses gives the direction.

Proof. Both sides satisfy Lemma 2 under the two-sided null, so their average does; the inclusion is immediate since both statistics are nonnegative. ▫

Corollary C (restarts after alarms). Suppose the monitor restarts after every alarm, with a new truncation level τ_k and baseline $m_0^{(k)}$ for segment k that are measurable at the start of the segment, and that the two-sided null holds in every segment relative to that segment's baseline; with $b = 0$ this means that $m_0^{(k)}$ equals the true conditional mean of $\min(r_t^2, \tau_k^2)$ throughout segment k. Let N_m be the number of alarms within the first m monitored steps (calibration steps are not counted). Then $\mathbb{E}[N_m] \le m/A$.

Proof sketch. Let W_t be the sum of $\bar{M}$ at all past alarms, plus the current $\bar{M}$, minus the number of monitored steps so far. By Lemma 2 applied in each segment, W is a supermartingale with $W_0 = 0$ (at an alarm the statistic moves into the sum unchanged, and calibration steps leave W unchanged). Each alarm adds at least A to the sum. Optional stopping at the bounded time $\rho \wedge n$, with ρ the time of the m-th monitored step, and monotone convergence as $n \to \infty$ give $A\,\mathbb{E}[N_m] - m \le 0$. ▫

Numerically (E4e) [SYNTHETIC]: with $A = 300$ the true baseline gave 26.5 alarms per 10,000 monitored steps, against the bound of 33.3. The streaming monitor with a plug-in baseline recalibrated on 390 steps gave 30.1; this is an empirical check only, since a plug-in baseline is not covered by the corollary (Corollary F below covers a confidence-sequence one). Corollary C is the bookkeeping behind "false alarms per day" over a long series; it is not a single anytime-valid statement for the whole series.

3.3 Why a decrease detector needs a bounded statistic

Proposition 2. Let $\mathcal{H}$ be the set of all probability laws on $[0, \infty)$ with mean at least 1. If $E: [0, \infty) \to [0, \infty]$ is measurable and $\mathbb{E}_P[E(Y)] \le 1$ for every $P \in \mathcal{H}$, then $E(y) \le 1$ for every $y \ge 0$.

Proof. For $y \ge 1$ the point mass δ_y lies in $\mathcal{H}$, so $E(y) \le 1$. For $y < 1$ and $\varepsilon \in (0, 1)$, let $z = (1 - (1 - \varepsilon)y)/\varepsilon$, so that $P_\varepsilon = (1 - \varepsilon)\delta_y + \varepsilon\delta_z$ has mean exactly 1 and lies in $\mathcal{H}$. Then $(1 - \varepsilon)E(y) + \varepsilon E(z) \le 1$, and since $E \ge 0$, $E(y) \le 1/(1 - \varepsilon)$. Let $\varepsilon \to 0$. ▫

The sequential version is what matters for monitoring.

Corollary 3 (no e-process for "the variance has not fallen"). Let $\mathscr{P}^{\downarrow}$ be the set of laws of real sequences $(r_t)_{t \ge 1}$ with $\mathbb{E}[r_t^2 \mid r_1, \ldots, r_{t-1}] \ge m_0$ for every t. Let $W_t = W_t(r_1, \ldots, r_t) \ge 0$ satisfy $\mathbb{E}_P[W_t] \le 1$ for every $P \in \mathscr{P}^{\downarrow}$ and every fixed t (a requirement weaker than being an e-process). Then $W_t \le 1$ for every t and every path. Consequently a Shiryaev-Roberts statistic built from such processes started at each time satisfies $M_t \le t$ on every path, so it cannot alarm before time A, whatever the data.

Proof. Fix t, a path $(x_1, \ldots, x_t)$ and $\varepsilon \in (0, 1)$. Build $P_\varepsilon \in \mathscr{P}^{\downarrow}$ step by step. If the past agrees with $(x_1, \ldots, x_{s-1})$ and $x_s^2 \ge m_0$, draw $r_s = x_s$; if $x_s^2 < m_0$, draw $r_s = x_s$ with probability $1 - \varepsilon$ and $r_s = z_s$ otherwise, with $z_s^2 = (m_0 - (1 - \varepsilon)x_s^2)/\varepsilon$, so that the conditional mean of r_s^2 is exactly m_0. Off the path, and after time t, draw $r_s = \sqrt{m_0}$. Then $P_\varepsilon(r_1 = x_1, \ldots, r_t = x_t) \ge (1 - \varepsilon)^t$ and, since $W_t \ge 0$, $1 \ge \mathbb{E}_{P_\varepsilon}[W_t] \ge (1 - \varepsilon)^t\, W_t(x_1, \ldots, x_t)$. Let $\varepsilon \to 0$. For the last claim, each started process is at most 1, so $M_t \le t$. ▫

This is a one-sided relative of the classical result of Bahadur and Savage (1956), who showed that without restrictions on the tails the mean of a distribution admits no nontrivial test. The asymmetry is what matters here. The *increase* null "mean at most 1" on $[0, \infty)$ does admit nontrivial e-variables, for example $E(y) = y$, which is why an increase detector works on raw squared returns (Section 5). A fall in the variance itself, however,

cannot be detected by any nonparametric e-test, whatever statistic it computes. A decrease detector must therefore change its target to a bounded functional of the returns. For a statistic with values in $[0, B]$ the null "conditional mean at least 1" does admit nontrivial e-variables, namely the bets $1 + \lambda(1 - y)$ with $\lambda \in [0, 1/(B - 1)]$, and up to domination these are the only ones (Lemma 5 in Section 4). Truncation is one way to obtain a bounded functional, and the truncated variance $\mathbb{E}[\min(r^2, \tau^2)]$ falls exactly when σ falls under the fixed-shape assumption of Lemma 1. It is not the only way: exceedance indicators $1\{|r_t| > q\}$, ranks, or any bounded increasing transform of r_t^2 would also do (rank-based detectors, exactly valid for exchangeable returns, are studied in ALT5 and E14). Truncation is the one with a direct volatility meaning and a known bias (Section 2). The price is a speed limit: one step can multiply the wealth of a decrease bet by at most $B/(B - 1)$ (Corollary 7). For increases, clipping is not needed for validity, but it bounds the effect of a single jump on the wealth, $f_t^{\uparrow} \le 1 + \lambda(B - 1)$, which is the betting counterpart of the bounded influence of the clipped estimator.

3.4 An estimated baseline

The guarantees above assume that the baseline m_0 and the level τ are fixed in advance, or at least that the null holds with whatever values the monitor uses. In practice both are estimated from a calibration window. The *plug-in* baseline (the mean of X over the window, with τ from the same window) is not covered, and its effect is measurable [SYNTHETIC]:

- With 390 calibration steps, the one-day false alarm rate of the increase side was 4.58%, 4.68% and 5.10% (Gaussian, Student t_3, jumps), against 1.75%, 1.73% and 1.62% with the true baseline; with 1440 steps it was 2.75%, 2.55% and 2.00% (4000 paths, Monte Carlo standard error 0.34 points, E4e). The plug-in roughly triples the rate at short calibration, and only the slack of Ville's inequality keeps it near the bound.
- The excess grows with the horizon, because a baseline that is slightly too low gives the log wealth a small positive drift that the bets eventually find. With exactly i.i.d. standardised returns and a one-day plug-in baseline, the false alarm rate over five days was 44% (95% interval 39% to 50%), where a known baseline would guarantee at most 25% (Theorem B; E10, Q10b2). Over seven days at band 0 it was 51% to 57% for Gaussian, Student t_4 and jump returns, where a known baseline would guarantee at most 35% (ALT5).
- In 30-day streams with no change and recalibration after every alarm, the plug-in raised 0.094 (Gaussian) and 0.084 (Student t_3) false alarms per 1440-step day against a nominal 0.05, with 95% lower limits above 0.05 (E14, Q14h). The accompanying theory note on running truncation levels shows why, at a fixed calibration length n. Let T_A be the first alarm of the two-sided plug-in rule, and for $\eta > 0$ and $\kappa < \infty$ let $p_{\eta,\kappa}$ be the probability that the calibration window gives the best bet of the detector, on either side, an expected log growth of at least η per monitored step under the null, with a largest factor of at most e^{κ}; $p_{\eta,\kappa}$ depends on n and on the law of the returns but not on A. Proposition A.6 of the note shows that $\liminf_{A \to \infty} \mathbb{P}(T_A \le m_A) \ge p_{\eta,\kappa}$ for $m_A = \lceil (\log A)^2 \rceil$, and Lemma A.7 proves $p_{\eta,\kappa} > 0$ for some η and κ under all four laws of E14 (Gaussian, Student t_3, jumps and price ticks). Since $m_A/A \to 0$, no bound $\mathbb{P}(T_A \le m) \le C\,m/A$ with a constant C holds for all A and m. The lemma uses calibration windows that are far calmer than the law they come

from, and therefore extremely rare, so it says nothing about how often a positive drift occurs; the share of calibration windows with a positive drift is a numerical estimate, 77% to 83% at 390 calibration steps (E14, Q14w) [SYNTHETIC]. Nor does the proposition account for the size of the failure at $A = 7800$: its own lower bound on the five-day rate is 0.5% to 0.8% at 390 calibration steps, against the 40.6% to 42.9% simulated (E14, Q14a).

A confidence sequence restores a proof. The construction uses a sample split, which is needed because a τ computed from the whole window makes each X_i in the window depend on later returns.

Construction. The first n_τ returns of the calibration window fix $\tau = c\hat{s}$, so τ is $\mathcal{F}_{n_\tau}$-measurable (the default uses 25% of the window). For the later returns, indexed $i = 1, 2, \ldots$, let $Z_i = \min(r_i^2, \tau^2)/\tau^2 \in [0,1]$ and $\eta_i = \mathbb{E}[Z_i \mid \mathcal{F}_{i-1}]$. Choose predictably a centre $\hat{z}_{i-1} \in (0,1)$ and bets $\lambda_i^{\uparrow}, \lambda_i^{\downarrow} > 0$ with $\lambda_i^{\uparrow}(1 - \hat{z}_{i-1}) \le c_0$ and $\lambda_i^{\downarrow}\hat{z}_{i-1} \le c_0$ for a fixed $c_0 \in (0,1)$. With $\psi_a(\lambda) = (-\log(1-\lambda a) - \lambda a)/a^2$, $e_i = (Z_i - \hat{z}_{i-1})^2$, $\psi_i^{\uparrow} = \psi_{1-\hat{z}_{i-1}}(\lambda_i^{\uparrow})$, $\psi_i^{\downarrow} = \psi_{\hat{z}_{i-1}}(\lambda_i^{\downarrow})$ and error levels $\delta_u, \delta_l \in (0,1)$, put

$$U_t = \frac{\sum_{i \le t} \lambda_i^{\uparrow} Z_i + \log(1/\delta_u) + \sum_{i \le t} \psi_i^{\uparrow} e_i}{\sum_{i \le t} \lambda_i^{\uparrow}}, \qquad L_t = \frac{\sum_{i \le t} \lambda_i^{\downarrow} Z_i - \log(1/\delta_l) - \sum_{i \le t} \psi_i^{\downarrow} e_i}{\sum_{i \le t} \lambda_i^{\downarrow}}.$$

They bound the bet-weighted averages $\bar{\eta}_t^{\uparrow} = \sum_{i \le t} \lambda_i^{\uparrow}\eta_i / \sum_{i \le t} \lambda_i^{\uparrow}$ and $\bar{\eta}_t^{\downarrow}$ (defined in the same way with $\lambda^{\downarrow}$) at all times at once: $\mathbb{P}(\exists t : \bar{\eta}_t^{\uparrow} \ge U_t) \le \delta_u$ and $\mathbb{P}(\exists t : \bar{\eta}_t^{\downarrow} \le L_t) \le \delta_l$, with no assumption on dependence or on how η_i varies (Theorem D.1 in the appendix). With $1 - \hat{z} = \hat{z} = 1$ these are the predictable plug-in empirical Bernstein bounds of Waudby-Smith and Ramdas (2024, Theorem 2); the one-sided ranges let the lower bound use bets up to $c_0/\hat{z}$, which matters because $\hat{z} \approx 0.11$ for a clipped squared Gaussian return.

Monitoring rule. Calibration ends at a stopping time $T_0 > n_\tau$ (a fixed length, or a rule that stops when the relative width of the bounds falls below a target). Set $m^{\uparrow} = (1+b)\tau^2 \min(U_{T_0}, 1)$ and $m^{\downarrow} = (1-b)\tau^2 \max(L_{T_0}, 0)$, run the two Shiryaev-Roberts statistics of Section 3.2 from T_0 with these baselines and bets fixed at T_0 (the decrease side is switched off when $m^{\downarrow} = 0$ or $\tau^2/m^{\downarrow} \le 1$), and alarm at the first $t > T_0$ with $\max(M_t^{\uparrow}, M_t^{\downarrow}) \ge 2A$.

Theorem D (guarantee with an estimated baseline). Let $\delta = \delta_u + \delta_l$ and assume

$$(H^{\uparrow})\ \mu_t \le (1+b)\tau^2 \bar{\eta}_{T_0}^{\uparrow} \quad \text{and} \quad (H^{\downarrow})\ \mu_t \ge (1-b)\tau^2 \bar{\eta}_{T_0}^{\downarrow} \qquad \text{for all } t > T_0,$$

with nothing assumed about the calibration window beyond $0 \le X \le \tau^2$. Then

1. for every integer $m \ge 1$, $\mathbb{P}(\text{an alarm at one of } T_0 + 1, \ldots, T_0 + m) \le \delta + m/A$, and more sharply $\le \delta + (1-\delta)m/A$ when $m < A$;
2. if $T_0 < \infty$ almost surely, the expected number of monitored steps before the first alarm is at least $(1-\delta)A$;
3. one side alone at level A satisfies the bound with δ_u (increase, under $H^{\uparrow}$) or δ_l (decrease, under $H^{\downarrow}$) in place of δ.

The hypotheses refer to bet-weighted averages, whose weights the algorithm chooses from the data, so whether they hold can depend on the bets. A sufficient condition that does not depend on the algorithm: since $\tau^2\eta_i = \mu_i$ for $i > n_\tau$ and a weighted average lies between the smallest and the largest term, $(H^{\uparrow})$ holds for every bet sequence when $\sup_{t > T_0} \mu_t \le (1+b)\min_{n_\tau < i \le T_0} \mu_i$, and $(H^{\downarrow})$ when $\inf_{t > T_0} \mu_t \ge (1-b)\max_{n_\tau < i \le T_0} \mu_i$. Missing

returns during monitoring are handled as in Section 2, and m counts every monitored step.

Proof sketch (full proof in Appendix A). Let G be the event that both bounds hold at T_0; it is known at T_0, and because the coverage of Theorem D.1 is uniform in time, $\mathbb{P}(G^c) \le \delta$ for every stopping time T_0. On G, $(H^\uparrow)$ and $(H^\downarrow)$ imply $\mu_t \le m^\uparrow$ and $\mu_t \ge m^\downarrow$, so on $G \cap \{t > T_0\}$ both statistics satisfy Lemma 2. Applying Ville's inequality to $1_{G \cap \{T_0 = s\}}(\bar{M}_t + m - (t-s))$ for each s and summing over s gives $\mathbb{P}(G \cap \text{alarm within } m) \le \mathbb{P}(G)\, m/A$. Add $\mathbb{P}(G^c) \le \delta$. ▫

Corollary E (a constant conditional law). If from step $n_\tau + 1$ until the change the conditional law of r_t given $\mathcal{F}_{t-1}$ is one fixed law P_0 (for example i.i.d. returns with jumps and fat tails), then $\mu_t = \mathbb{E}_{P_0}[\min(r^2, \tau^2)]$ is constant, $(H^\uparrow)$ and $(H^\downarrow)$ hold with $b = 0$, and with no change $\mathbb{P}(\text{false alarm within } m \text{ monitored steps}) \le \delta + m/A$ with the baseline estimated from the data.

Corollary F (recalibration after every alarm). If every segment has its own split, bounds and error budget δ, and $(H^\uparrow)$, $(H^\downarrow)$ hold in every segment relative to its own calibration averages, then the number N_m of alarms within the first m monitored steps satisfies $\mathbb{E}[N_m] \le (\delta + m/A)/(1-\delta)$. With $\delta = 0$ this is Corollary C. (Proof in Appendix A.)

Choosing A and δ. For a target α per day of m steps, choose $\delta < \alpha$ and $A = m/(\alpha - \delta)$. With $\delta = 0.01$ and $\alpha = 5\%$, $A = 9750$ for a 390-step day and $A = 36,000$ for a 1440-minute day. Because the delay grows like $\log A$, the larger A costs little ($\log 9750/\log 7800 \approx 1.025$); the real cost is the width of the bounds (Section 5.3). Two remarks. Under volatility clustering the bounds still cover the bet-weighted average of the conditional means within the calibration window, but the monitoring null "never above that average" is then simply false (Section 5.4): the confidence sequence protects against estimation error, not against this model mismatch. And because the coverage is uniform in time, the calibration may end at any stopping time, which a fixed-n interval would not allow.

4 Detection delay

Section 3 controls false alarms. This section asks how fast the detector reacts to a real change, what robustness costs in speed, and what limits the speed of decrease detection. Proofs not given here are in Appendix B. Numerical statements are from E10 and E11 and are [SYNTHETIC].

4.1 Growth rate and information

Post-change model. From a change time ν on, $Y_\nu, Y_{\nu+1}, \ldots$ are i.i.d. with law Q and independent of $\mathcal{F}_{\nu-1}$. Write $V = Y - 1$ for the increase side and $V = 1 - Y$ for the decrease side; V takes values in $[v_{\text{lo}}, v_{\text{hi}}]$, equal to $[-1, B-1]$ or $[1-B, 1]$, and the admissible bets are $\lambda \in [0, \lambda_{\max})$ with $\lambda_{\max} = 1/|v_{\text{lo}}|$. A change means $\mathbb{E}_Q[V] > 0$. The worst-case delay of a stopping rule T is Lorden's (1971)

$$\text{WADD}(T) = \sup_{\nu \ge 1} \operatorname{ess\,sup} \mathbb{E}_\nu[(T - \nu + 1)^+ \mid \mathcal{F}_{\nu-1}].$$

The *growth rate* of a bet is $g(\lambda) = \mathbb{E}_Q[\log(1 + \lambda V)]$. It is strictly concave with $g(0) = 0$ and $g'(0) = \mathbb{E}_Q[V] > 0$, so it has a unique maximiser $\lambda^*(Q)$ in $[0, \lambda_{\max}]$. Let $\mathcal{H} = \{P \text{ on } [0,B] : \mathbb{E}_P[V] \le 0\}$ be the nonparametric null class of one step, and

$$\text{KL}_{\text{inf}}(Q) = \inf_{P \in \mathcal{H}} \text{KL}(Q \| P).$$

Lemma 4 (growth is bounded by information). If $E \geq 0$ and $\mathbb{E}_P[E(Y)] \leq 1$ for a law P, then $\mathbb{E}_Q[\log E(Y)] \leq \mathrm{KL}(Q \| P)$.

Lemma 5 (the best nonparametric growth). Let $B < \infty$. (a) Every e-variable for the whole class $\mathcal{H}$ is dominated by a bet: there is $\lambda \in [0, \lambda_{\max}]$ with $E(y) \leq 1 + \lambda v(y)$ for all y. (b) If $0 < \lambda^* < \lambda_{\max}$, then $\max_\lambda g(\lambda) = g(\lambda^*) = \mathrm{KL}_{\inf}(Q)$, and the infimum is attained by $P^*(dy) = Q(dy)/(1 + \lambda^* v(y))$.

Part (b) is the dual form, for a bounded statistic, of the information number $K_{\inf}$ of Honda and Takemura (2010), and P^* is the reverse information projection of Q onto the null (Grünwald, de Heide and Koolen 2024; Larsson, Ramdas and Ruf 2024). Since every bet is an e-variable for the true pre-change law P_0 (when $P_0 \in \mathcal{H}$), and since X and Y are functions of r, Lemmas 4 and 5 and the data processing inequality give:

Corollary 6 (information ladder). With $g_{\mathrm{grid}} = \max_k g(\lambda_k)$,

$$\mathrm{KL}(Q_r \| P_{0,r}) \geq \mathrm{KL}(Q_X \| P_{0,X}) \geq \mathrm{KL}_{\inf}(Q) \geq g_{\mathrm{grid}}.$$

The three ratios between neighbouring terms are the *truncation loss*, the *nonparametric loss* (the price of a null that restricts only the mean of Y, which is what makes the detector valid for every return law) and the *grid loss*. The ratio $\mathrm{KL}(Q_r \| P_{0,r})/g_{\mathrm{grid}}$ is the first-order delay ratio of the robust detector against the likelihood-ratio detector that knows both laws.

Corollary 7 (speed limits). Every bet satisfies $f_\lambda \leq 1 + \lambda_{\max} v_{\mathrm{hi}}$, so by Lemma 5(a) every e-variable for $\mathcal{H}$ satisfies $\log E \leq \log B$ on the increase side and $\log E \leq \log(B/(B-1))$ on the decrease side, and $\mathrm{KL}_{\inf}(Q)$ obeys the same bounds however large the change. For the detector itself the statement is deterministic: $M_n \leq \sum_k w_k \sum_{j=1}^{n} F_k^{n-j+1}$ with $F_k = 1 + \lambda_k v_{\mathrm{hi}}$, so T_A is never smaller than the first n at which this sum reaches A.

With $c = 3$ and Gaussian returns, $\log(B/(B-1)) = 0.117$ nats per step. This per-step cap holds for every valid bet; how many steps it implies depends on the bets. For the default decrease grid at $A = 7800$ the floor is 36, 73, 122 and 180 steps for $c = 2, 3, 4$ and 5, and a complete collapse of volatility (all returns zero) is detected after exactly 73 steps with $c = 3$ (E11). A single bet close to $\lambda_{\max}$ does better: with F close to $B/(B-1)$ the deterministic bound reads $M_n \leq B(F^n - 1)$, so no Shiryaev-Roberts e-detector whose bets are valid over the whole null can alarm before $n = \log(1 + A/B)/\log(B/(B-1))$, which is 57.7 at $A = 7800$ and 63.6 at the two-sided level $2A$, and a single bet just below $\lambda_{\max}$ alarms after 58 (64) steps when all returns are zero. The corresponding floors for any valid bets at $A = 7800$ are 29, 96 and 141 steps for $c = 2, 4$ and 5 (E11). For rules that are not e-detectors the only lower bound we have is the asymptotic one of Theorem 9(a). This is the quantitative form of the impossibility of Section 3.3: the bound B that makes a decrease detectable also caps the evidence that one step can provide.

4.2 Delay bounds and first-order optimality

Theorem 8 (non-asymptotic delay bound). Under the post-change model, for the Shiryaev-Roberts mixture or its CUSUM-type version:

a. for every A, every change time v and every pre-change history,

$$\mathbb{E}[(T_A - v + 1)^+ \mid \mathcal{F}_{v-1}] \leq \min_{k:\, g_k > 0} \frac{\log(A/w_k) + o_k}{g_k},$$

where $g_k = g(\lambda_k)$, $Z_k = \log(1 + \lambda_k V)$ with V drawn from Q, and $o_k = \min(\sup Z_k, \mathbb{E}[(Z_k^+)^2]/\mathbb{E}[Z_k])$ bounds the mean overshoot;

b. $\mathrm{WADD}(T_A) = \mathbb{E}_Q[T_A^0]$, where T_A^0 is the alarm time of the detector started from $M_0 = 0$ on i.i.d. data from Q: the worst case is a change at the first monitored step.

Proof. (a) For $t \geq v$ all terms of the Shiryaev-Roberts sum are nonnegative, so $M_t \geq w_k L_t^{(v,k)}$. Let $S_n = \sum_{i=0}^{n-1} Z_{k,v+i}$ and $N = \inf\{n \geq 1 : S_n \geq \log(A/w_k)\}$; then $(T_A - v + 1)^+ \leq N$. N depends only on post-change data, which are independent of $\mathcal{F}_{v-1}$. S is a random walk with drift $g_k > 0$ and bounded increments, so Wald's identity gives $g_k\,\mathbb{E}[N] = \log(A/w_k) + \mathbb{E}[S_N - \log(A/w_k)]$, and the overshoot is bounded by $\sup Z_k$ and, in mean, by Lorden's (1970) inequality. (b) Each recursion $x \mapsto f(x+1)$ (or $f\max(x, 1)$) is nondecreasing in x, so from any state $M_{v-1} \geq 0$ the statistic after v dominates the one started from 0 on the same post-change data. ▫

Part (a) is the elementary version, for linear bets on a bounded statistic, of the delay bounds that Shin, Ramdas and Rinaldo (2024) give for mixtures of exponential baseline e-detectors. It pays the full $\log(1/w_k) = \log K$ for not knowing the right bet and ignores the help of the other bets and start times; in simulation it is 1.3 to 2.7 times the true delay (E11). Part (b) means that simulations with the change at the first monitored step measure the worst-case delay exactly, and any pre-change history can only shorten it.

Theorem 9 (first-order asymptotics). Fix Q with $\mathbb{E}_Q[V] > 0$ and an interior maximiser $0 < \lambda^* < \lambda_{\max}$, so that $g(\lambda^*) = \mathrm{KL}_{\inf}(Q)$. As $A \to \infty$:

a. *Lower bound.* Let T be any stopping rule based on the returns whose average run length is at least A under every i.i.d. return law P with $\mathbb{E}_P[\min(r^2, \tau^2)] \leq m^{\uparrow}$ (increase side; symmetrically for the decrease side). Then $\mathrm{WADD}_Q(T) \geq (\log A/\mathrm{KL}_{\inf}(Q))(1 + o(1))$.

b. *Fixed grid.* For a fixed grid with $g_{\mathrm{grid}} > 0$, $\mathrm{WADD}(T_A)/(\log A/\mathrm{KL}_{\inf}) \to \mathrm{KL}_{\inf}/g_{\mathrm{grid}} \geq 1$.

c. *Refined grid.* Fix $0 < a < b' < \lambda_{\max}$ with $\lambda^* \in [a, b']$, and use $K_A = 1 + \lceil (b' - a)\sqrt{\log A} \rceil$ equally spaced bets on $[a, b']$ with equal weights. Then $\mathrm{WADD}(T_A) \leq (\log A + \frac{1}{2}\log\log A + O(1))/\mathrm{KL}_{\inf}$.

d. *Continuous mixture.* For a mixing law on $[0, b']$ with a density at least $p_0 > 0$ on $[a, b']$ and $\lambda^* \in (a, b')$, $\mathrm{WADD}(T_A) \leq (\log A + \log\log A + O(1))/\mathrm{KL}_{\inf}$.

What is proved is a first-order lower bound in (a) and upper bounds in (b) to (d); the second-order terms in (c) and (d) are upper bounds, not exact rates. Together, (a) and (c) say that the delay of a refined grid divided by $\log A/\mathrm{KL}_{\inf}$ tends to 1, so such a grid is first-order asymptotically optimal among all rules that control the run length over the nonparametric null. A fixed grid is not: its delay ratio tends to $\mathrm{KL}_{\inf}/g_{\mathrm{grid}}$, which is 1.00 to 1.08 for the default grid in the cases of E11 (E11, Q11e) [SYNTHETIC], and at practical alarm levels its delays are in fact below $\log A/\mathrm{KL}_{\inf}$ (Section 4.3).

Proof idea (full proof in Appendix B). For (a), the law $P_r^*(dr) = Q_r(dr)/(1 + \lambda^* v(r))$ satisfies the null with equality and has $\mathrm{KL}(Q_r \| P_r^*) = \mathrm{KL}_{\inf}(Q)$, so Lorden's (1971) lower bound for a change from P_r^* to Q_r applies. Parts (b) to (d) combine Theorem 8(a) with a Taylor expansion of g around λ^*. ▫

First-order optimality for bounded means is known: Ram and Ramdas (2026, preprint) prove a universal lower bound of the form (a) for composite pre- and post-change classes and show that it is attained for bounded means, which covers the truncated statistic. Theorem 9 is an instance of their result, with explicit constants for linear bets on our statistic, and we make no claim of priority for the first-order statement. We give (a) to (d) because for linear bets on a bounded statistic the arguments are short, and because the comparison between fixed grids, refined grids and mixtures, with the rate at which a grid must be refined, is what matters for the choice of bets.

4.3 Second-order behaviour at practical alarm levels

The first-order formula $\log A/g$ is a poor guide at $A = 7800$: it overstates the delay of small changes by up to 112%, and the proved bound of Theorem 8(a) is 1.3 to 2.7 times the simulated delay. A second-order approximation from nonlinear renewal theory, $(\log A + \kappa - \zeta)/g$, with κ the mean overshoot and $\zeta \approx \log(1/g) + \gamma_E$ a memory term of the Shiryaev-Roberts statistic (γ_E is Euler's constant), was within 3% of the simulated mean delay of a single bet in all 20 cases of E11; we call it heuristic because we did not check its regularity conditions. The default grid costs 9% to 26% against the best single bet, and with the change after 500 pre-change steps, as in E4, delays are 17% to 34% shorter, because the statistic enters the change with a head start. The asymptotic regime arrives slowly: even at $A = 10^8$ the best single bet is 5% to 24% faster than $\log A/\mathrm{KL}_{\mathrm{inf}}$ (E11) [SYNTHETIC]. Supplement S1 gives the derivation, Table S1 and Figure S1.

4.4 The price of robustness and the truncation constant

Table 1 splits the first-order delay ratio against the Gaussian likelihood-ratio detector into the three factors of Corollary 6, for clean Gaussian data and $c = 3$, and compares it with simulated mean delays (E11, Q11b) [SYNTHETIC].

sd ratio	1.25	1.5	2	0.8	0.67	0.5
truncation loss	1.03	1.08	1.27	1.00	1.00	1.00
nonparametric loss	1.10	1.23	1.48	1.25	2.00	3.61
grid loss	1.02	1.02	1.04	1.02	1.05	1.05
predicted penalty $\mathrm{KL}/g_{\mathrm{grid}}$	1.16	1.37	1.95	1.27	2.09	3.79
simulated penalty against the Gaussian LR mixture	1.14	1.35	1.73	1.48	2.19	3.45

Table 1. The price of robustness on clean Gaussian data, $c = 3$ (E11). Penalties are ratios of mean delays, change at the first step, 3000 paths with common random numbers.

The information ratio explains the 10% to 70% delay penalty measured in E4 (Section 5.3). The largest factor is the nonparametric loss, not truncation. For decreases the nonparametric loss is the speed limit of Corollary 7: $\mathrm{KL}_{\mathrm{inf}}$ cannot exceed 0.117 nats per step, while the Gaussian information keeps growing with the size of the change.

Asymptotically, the robust detector keeps $\mathrm{KL}_{\mathrm{inf}}/\mathrm{KL} = 53\%$ to 88% of the speed of the Gaussian likelihood-ratio detector for increases (sd ratio 2 to 1.25) and 28% to 80% for decreases (0.5 to 0.8); under Student t_3 returns it keeps 94% to 98% of the speed of the correctly specified detector for increases, because heavy tails carry little information about scale (E11, Q11e). In exchange the guarantee of the robust detector covers every return law, while that of the Gaussian one holds only for Gaussian data (Section 5.1).

The truncation constant. For increases on clean data $c = 3$ costs 3% to 12% of delay against no truncation, and with jumps truncation is what keeps a burst of jumps from looking like a volatility increase: a burst moves the $c = 3$ detector only 3.5% as much as a real 25% increase, against 39% without truncation. For decreases the speed limit dominates and a smaller c is usually faster. A separate clip for the decrease side ($c_{\downarrow} = 2$, keeping $c = 3$ for increases) cut simulated median decrease delays by 13% to 51% for Gaussian, Student t_4 and jump returns, and by 32% to 52% for prices rounded to ticks of 0.25 and 0.75 standard deviations, but was 12% slower for a 20% fall at a tick of 0.5 standard deviations (paired difference of 29 steps, standard error 1.7, slower on 66% of paths); on the five coins, which have few zero returns, it was neutral to slightly positive (E11f) [SYNTHETIC and REAL]. Its false alarm guarantee holds with a known baseline or with the confidence-sequence baseline, which the monitor supports for the split (each side at level $\delta/2$): each side is then a valid e-detector for its own truncated statistic, and the two-sided rule of Proposition 1 needs only each side's guarantee. The simulations used the plug-in baseline, whose one-day false alarm rates (5.0% to 6.7%, and 5.0% to 6.05% without the split) are empirical only. Details are in Supplement S2.

4.5 Clustering, forecast-relative monitoring and a signal cap

Under persistent volatility clustering the conditional mean of X_t wanders even when the unconditional variance is constant, so the nulls of Section 3 are false and the raw monitor alarms on most days (Section 5.4). A natural fix is to monitor $z_t = r_t/\hat{\sigma}_t$, with $\hat{\sigma}_t^2$ a predictable volatility forecast; the null then reads "the forecast is right up to a constant". The following exact calculation shows what this costs when the forecast model is GARCH(1,1) (Bollerslev 1986).

Proposition 10 (signal cap). Let $\alpha, \beta > 0$ with $\alpha + \beta < 1$ and $\omega > 0$. Before a break the returns follow GARCH(ω, α, β), and the forecast is the pre-break filter $s_t = \omega + \alpha r_{t-1}^2 + \beta s_{t-1}$, run without change after the break. From the break on, the returns follow GARCH($k\omega, \alpha, \beta$) with $k > 0$: $r_t = \sigma_t \varepsilon_t$, $\sigma_t^2 = k\omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$, with i.i.d. ε_t of mean 0 and variance 1. Then $d_t = \sigma_t^2 - s_t$ satisfies $d_t = (k-1)\omega + \beta d_{t-1}$ deterministically, so $d_t \to (k-1)\omega/(1-\beta)$ geometrically at rate β, and in the post-break stationary regime

$$\rho(k) := \frac{\mathbb{E}[r_t^2]}{\mathbb{E}[s_t]} = \frac{k(1-\beta)}{1-\alpha-\beta+\alpha k},$$

which does not depend on ω, increases in k, and satisfies $\rho(k) < (1-\beta)/\alpha$ for every k, however large the break.

Proof. Subtracting the two recursions, the terms αr_{t-1}^2 cancel, which gives the recursion for d_t. In the post-break stationary regime $\mathbb{E}[r_t^2] = \mathbb{E}[\sigma_t^2] = k\omega/(1-\alpha-\beta)$ and $\mathbb{E}[s_t] = \mathbb{E}[\sigma_t^2] - (k-1)\omega/(1-\beta)$. Dividing and simplifying gives $\rho(k)$. Its derivative in k has the sign of $(1-\beta)(1-\alpha-\beta) > 0$, and $\rho(k) \to (1-\beta)/\alpha$ as $k \to \infty$. ▫

Two exact facts sit behind this. A scale break ($r_t \mapsto \sqrt{k}\, r_t$) and an intercept break ($\omega \mapsto k\omega$) lead to the same post-break model and differ only in the transient right after the break. And the filter absorbs most of the break through its αr^2 term: at persistence $\alpha + \beta = 0.99$ with $\alpha = 0.05$, a doubling of the long-run variance appears after standardisation as an increase of the ratio of means to $\rho(2) = 1.091$, and no break can push it above $(1 - \beta)/\alpha = 1.2$. Here $(1 - \beta)/\alpha$ caps the post-break ratio of the mean squared return to the mean forecast; it is not a fraction of the shift. The monitor sees the mean of the ratio, $\mathbb{E}[r_t^2/s_t]$, which is a little larger by Jensen's inequality (1.116 for $k = 2$ in simulation) and which stayed below 1.28 for k up to 1000 at this persistence; we have a proof of the cap only for the ratio of means. Figure 1 compares the closed form with simulation: the simulated ratio of means, pooled over paths, agrees with $\rho(k)$ within two standard errors at every point (E10, Q10d) [SYNTHETIC]. (An average of per-path ratios is biased upward when the path means are noisy; at persistence 0.999 it exceeded the cap, which is why we pool.)

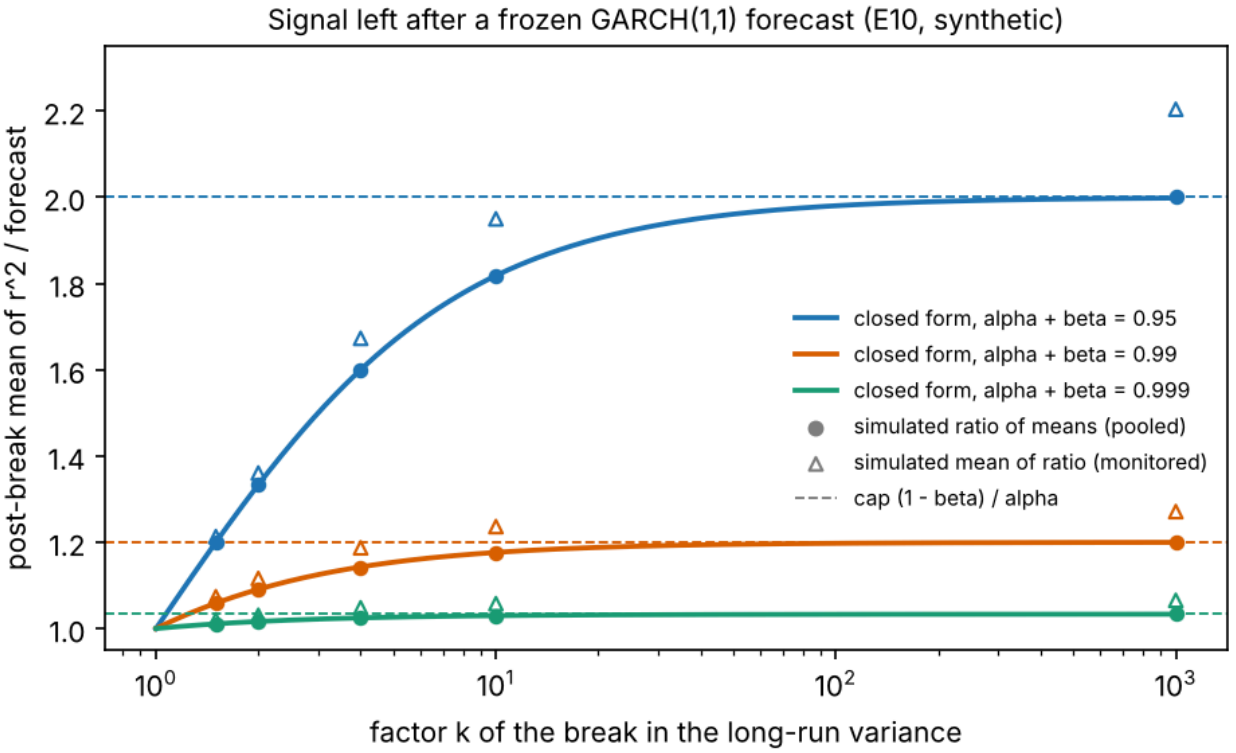


Figure 1. The ratio of means $\rho(k)$ (lines, closed form; filled circles, simulation pooled over paths, with bars of two bootstrap standard errors) and the mean of the ratio that the monitor sees (open triangles), with the caps $(1 - \beta)/\alpha$ (dashed), for three persistences. 200 paths of 18,000 post-break steps per point; all standard errors are below 0.002 (E10, Q10d) [SYNTHETIC].

An information rate. With Gaussian innovations the per-step Kullback-Leibler divergence between the post-break and pre-break conditional laws of r_t is $\frac{1}{2}(q_t - 1 - \ln q_t)$ with $q_t = \sigma_t^2/s_t$. Let I be its average over the post-break path. By Lai (1998), any detector whose average run length to a false alarm is at least A needs, as $A \to \infty$ and under his condition that the post-change log-likelihood ratio grows at rate I uniformly in the change time, a worst-case delay of at least $(\ln A/I)(1 + o(1))$. At $A = 28,800$ this is an order-of-magnitude guide, not an exact bound. Table 2 gives the numbers.

$\alpha + \beta$	k	$\rho(k)$	**simulated mean of ratio**	I **(nats per step)**	$\ln A/I$ **(days of 1440 steps)**
0.95	2	1.333	1.361	0.0276	0.26

$\alpha+\beta$	k	$\rho\,k$	**simulated mean of ratio**	I **(nats per step)**	$\ln A/I$ **(days of 1440 steps)**
0.99	1.5	1.059	1.074	0.0015	4.7
0.99	2	1.091	1.116	0.0037	1.9
0.99	4	1.143	1.187	0.0093	0.77
0.99	1000	1.200	1.271	0.0187	0.38
0.999	2	1.016	1.031	0.0003	21

Table 2. Signal left after a frozen GARCH(1,1) forecast and the information rate of a break (E10, Q10d) [SYNTHETIC]. $A = 28,800$.

At persistence 0.99 no detector with this false alarm budget can be expected to see a doubling of the long-run variance much faster than within about two days, and at persistence 0.999 it is practically invisible for weeks. This is a property of the question, not of the monitor: a highly persistent volatility model can explain a level shift as its own slow wandering, the mirror image of the known finding that unmodelled level shifts make fitted GARCH models look nearly integrated (Diebold 1986; Lamoureux and Lastrapes 1990; Mikosch and Starica 2004). The observed median delays of the forecast-relative monitors (1523 to 1791 steps for a doubling, Section 5.4) are about 0.6 times the step at which the cumulative information reaches $\ln A$; the memory term of Section 4.3 can explain a factor of this size on its own (in an i.i.d. version of the same steady state with the true baseline the detector alarms at 0.41 to 0.58 times $\ln A/\mathrm{KL}$, E11), but we did not rerun the GARCH setting to separate it from the effect of the plug-in baseline.

5 Simulation evidence

All results in this section are **[SYNTHETIC]**: returns are simulated in per-step units with fixed seeds, and the settings of each experiment are collected in Appendix C. Unless stated otherwise the monitor uses $c = 3$ with the scaled MAD, the default grids of Section 3.2 and $A = 7800$, so that with a known baseline Theorem B bounds the probability of a false alarm within one 390-step day by 5%. “Jumps” adds to Gaussian returns, with probability 0.005 per step, an independent normal jump whose standard deviation is 8 times (E4, E8, E11) or 10 times (E9) the diffusive one. Student t_ν returns are scaled to unit variance. “Price ticks” are Gaussian returns rounded to multiples of half a standard deviation (about 20% zero returns).

5.1 Validity where classical detectors fail

E4 compares the two sides of the robust monitor with an e-detector on raw squared returns, a Shiryaev-Roberts mixture of Gaussian likelihood ratios for standard deviation ratios 1.25, 1.5 and 2 (increase side) at the same A, Page’s (1954) CUSUM on r_t^2 with its threshold calibrated by simulation to an average run length of about 7800 on Gaussian data, and a two-sided z-test of the variance over the last 60 steps at level 0.05, repeated at every step. The baseline of every method is estimated from a 2000-step calibration window drawn from the same law as the monitored data, so jumps and fat tails are part of “normal”.

Detector	Gaussian	jumps	Student t_5	price noise	slow SV
Robust e-detector, increase	1.7%	2.3%	0.7%	3.7%	26.3%
Robust e-detector, decrease	0.7%	0.7%	0.3%	0.3%	27.0%
E-detector on raw r_t^2, increase	1.7%	2.3%	1.3%	3.0%	26.3%
Gaussian likelihood-ratio e-detector, increase	2.7%	**70.7%**	**96.3%**	6.3%	29.0%
Page CUSUM, calibrated on Gaussian data	5.0%	**70.3%**	**97.3%**	9.7%	32.3%
Repeated 60-step z-test	**70.7%**	**90.0%**	**100%**	**78.0%**	**95.0%**

Table 3. Share of paths with a false alarm within one 390-step day, no change, 300 paths per cell (Monte Carlo standard error about 0.8 points at 2% and 1.3 points at 5%). "Price noise" adds i.i.d. noise with standard deviation 0.5 per-step return standard deviations to log prices; "slow SV" is a log-normal stochastic volatility with log-sd 0.15 and half-life 390 steps (E4, Q4a) [SYNTHETIC].

In the three columns where the null holds, all e-detectors stayed below 5%, including the increase detector on raw squared returns, as Section 3.3 predicts. These rates are measured: with a baseline estimated from 2000 steps no proof covers them (Section 3.4), and Section 5.2 gives the rates with the true baseline, to which the theorems apply. The Gaussian likelihood-ratio and CUSUM detectors, whose thresholds are correct only for Gaussian data, alarm on 70% to 97% of days under jumps or fat tails, and the repeated z-test fails everywhere. The last two columns are not cases where the null holds. Price noise makes the conditional mean of r_t^2 depend on the previous return (Section 5.4), so the 3.7% shows robustness to a mild violation over one day. Under slow stochastic volatility the conditional variance genuinely drifts, and every method alarms on a quarter or more of days.

5.2 How tight is the bound

E9 runs the two-sided monitor with the true baseline (m_0 from 20 million draws, τ floored at three ticks for the tick law) on 4000 paths per law (Table 4, Figure 2).

Horizon m (steps)	bound m/A	Gaussian	Student t_3	jumps	price ticks
98	1.3%	0.00%	0.18%	0.12%	0.05%
390	5.0%	1.1%	1.4%	1.3%	1.4%
1950	25%	12.8%	14.3%	13.2%	13.7%
3900	50%	29.6%	30.6%	29.7%	31.2%
7800	100%	55.9%	55.9%	56.4%	56.9%

Table 4. False alarm rate within m steps with the true baseline, two-sided rule, $A = 7800$ (E9, Q9a and Q9b) [SYNTHETIC]. Student t_3 has an infinite fourth moment. 95% intervals at one day are about ± 0.3 points.

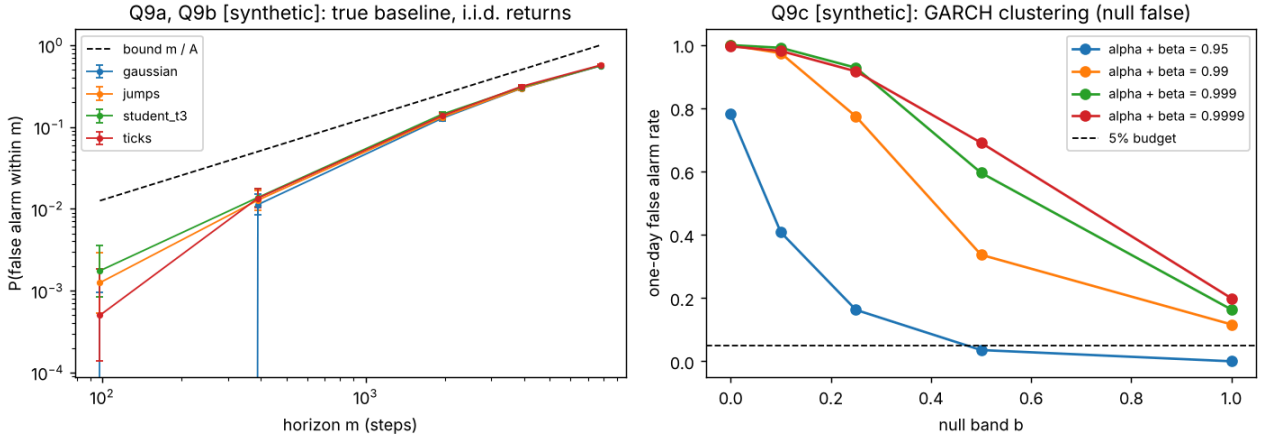


Figure 2. Left: false alarm rate within m steps against the bound m/A for four return laws, true baseline. Right: one-day false alarm rate under GARCH(1,1) clustering, where the null is false, as a function of the band b (E9) [SYNTHETIC].

The bound holds for every law, and the four laws give almost the same rates, which is what a nonparametric guarantee should look like once the statistic is bounded. It is conservative by a factor of about 4 at one day and about 2 at $m = A$. An exact optional stopping identity traces the slack to overshoot (a factor of about 1.25) and to expected wealth stranded on paths that never alarm (a factor of about 4), most of it held by the decrease side and the smallest increase bets. The bound itself is nearly sharp: for a single bet of 0.5, dynamic programming over the whole nonparametric null finds a law that reaches 95% of it; for the default mixture the worst law found reaches 42%, and a certified sharper bound is open (E11, Q11d; Supplement S2).

5.3 Detection delay, bet design and the cost of an estimated baseline

Against the benchmarks. Table 5 reports median delays after a change 500 steps into monitoring (E4, Q4b).

Detector	Gaussian	jumps	Student t_5
Increases, sd ratio 1.25, 1.5, 2			
Robust e-detector (default grid)	87, 35, 17 ($\leq$ 4.3%)	102, 36, 18 ($\leq$ 4%)	122, 49, 22 (3%)
Robust e-detector, CUSUM type	154, 51, 23 ($\leq$ 1%)	171, 52, 23 ($\leq$ 0.3%)	215, 72, 32 ($\leq$ 0.3%)
E-detector on raw r_t^2	84, 33, 15 ($\leq$ 4.3%)	246, 60, 23 ($\leq$ 5.3%)	134, 51, 21 ($\leq$ 4%)
Gaussian likelihood-ratio e-detector	78, 26, 10 ($\leq$ 5.3%)	61, 20, 9 (**74 to 78%**)	46, 10, 7 (**96 to 98%**)
Page CUSUM	64, 24, 10 (6 to 10%)	54, 20, 11 (**74 to 79%**)	50, 9, 6 (**97 to 99%**)
Decreases, sd ratio 0.5, 0.67, 0.8			
Robust e-detector (default grid)	71, 92, 147 ($\leq$ 2%)	72, 98, 160 (1%)	68, 96, 168 ($\leq$ 3%)

Detector	Gaussian	jumps	Student t_5
Gaussian likelihood-ratio e-detector	23, 46, 104 (6 to 9%)	23, 49, 121 (3 to 7%)	29, 81, 282 ($\leq$ 2%)

Table 5. Median delay in steps after a change at step 500 of monitoring; in brackets, the share of paths with a false alarm before the change. 300 paths per cell (E4, Q4b) [SYNTHETIC]. The decrease detector on raw r_t^2 does not exist (Section 3.3).

On clean Gaussian data the parametric detectors are faster, and the robust detector pays 10% to 70% extra delay for increases, the penalty that the information ladder of Table 1 predicts. With jumps or fat tails the parametric detectors alarm before the change on 74% to 99% of paths, so their delays have no meaning. Truncation matters for increases under jumps: at sd ratio 1.25 the median delay is 102 steps with truncation and 246 without. The CUSUM-type version of the robust detector is slower, as $C_t \leq M_t$ implies. For decreases the Gaussian likelihood-ratio detector was faster and close to its nominal false alarm rate in these designs, because jumps only add evidence against a decrease; this rests on normality, and real one-minute returns contain many exact zeros (5.7% of Bitstamp BTC/USD minutes, E5 [REAL]), which a Gaussian likelihood reads as evidence of low volatility.

Bet design. The quarticity estimator can enter the detector through the size of the bet, but a plug-in bet built from a local truncated quarticity was slower than the fixed grid in 14 of 18 cells, by up to 9% (E4, Q4c), because the quadratic approximation behind it makes it too small (E11, Q11a). A grid designed for robustness across four tail shapes was at least as fast as the default in all 48 validation cases, by about 5% in the median, while a grid tuned on Gaussian data was up to 55% slower under Student t_3 returns (E11, Q11c; Supplement S2).

The cost of an estimated baseline. E8 compares the plug-in baseline with the confidence-sequence baseline of Theorem D ($\delta = 0.01$ per side and $A = 9750$, so that $\delta + m/A = 5\%$ per day). Without a change the confidence-sequence monitor had at most 0.27% false alarms per day in nine settings (three laws, calibration of 390, 1440 and 5000 steps, 10,000 paths each), and the confidence bound failed on at most 0.08% of paths against an allowed 1% (E8, Q8a). The plug-in was at 4.5% to 4.9% on the increase side at 390 calibration steps, at the bound but not beyond it within Monte Carlo error at one day; Section 3.4 shows that it does exceed the bound over longer horizons. The price of the proof is delay (Table 6).

Returns, sd ratio	390 steps: plug-in / CS	1440 steps: plug-in / CS	5000 steps: plug-in / CS
Gaussian, 1.25	90 / 645 (76%)	88 / 222	88 / 134
Gaussian, 1.5	36 / 103	35 / 59	35 / 46
Gaussian, 2	17 / 31	17 / 24	17 / 20
Gaussian, 0.8	143 / 633 (85%)	137 / 269	139 / 199
Gaussian, 0.67	89 / 207	87 / 135	87 / 114
Gaussian, 0.5	64 / 122	65 / 92	64 / 80
Student t_5, 1.25	128 / 904 (35%)	131 / 391	130 / 219

Table 6. Median delay in steps after a change at step 500, plug-in against confidence-sequence (CS) baseline, by calibration length; in brackets, the share detected within 2500

steps when below 90%. 2000 paths per cell (E8, Q8b) [SYNTHETIC]. With 390 calibration steps the plug-in detects only 92% to 94% of increases in this sense, because it alarms before the change on 6% to 7.5% of paths.

The proof is expensive with short calibration and cheap with long calibration: at 390 steps the confidence-sequence monitor needs 1.8 to 7.6 times the plug-in delay over the 18 cells of E8 (the largest factor for jump returns at sd ratio 1.25), at 1440 steps about 1.4 times for a doubling or halving and 1.5 to 3.0 times for smaller changes, and at 5000 steps 1.2 to 1.7 times. Most of the width is intrinsic rather than a weakness of the bound: from 292 observations, the median upper bound sits 37% above the true mean for Gaussian returns, while the bound with the best constant bet for the true law (a benchmark that knows the distribution, not a limit for every betting bound) sits 29% above it and a normal approximation without a finite-sample guarantee 19% above it; at 1440 steps and more our upper bound is within 3.5 points of that benchmark (E8, Q8d). The truncated squared return has a standard deviation 1.4 to 1.6 times its mean, so a mean is hard to pin down from a few hundred minutes. The choice of δ matters little between 0.01 and 0.03 (delays within 5%, E8, Q8g), and a calibration that stops when the relative width of the bounds reaches a target behaves almost like a fixed length chosen for that width (E8, Q8c).

5.4 Clustering, forecasts and price noise: when the null is false

Volatility clustering. Under GARCH(1,1) returns with constant unconditional variance the conditional mean of X_t moves above and below m_0, so the nulls of Section 3 are false. With the true unconditional baseline, the monitor alarmed on 78% of 1440-step days at persistence 0.95 and on 100% at persistence 0.99 or more; a band $b = 0.5$ absorbs short-lived clustering, but even $b = 1$ leaves 12% to 20% of days with an alarm when the half-life is an hour or more (E9, Q9c; Figure 2, right). These alarms are not errors of the proof: persistent GARCH clusters are changes in conditional volatility, and the monitor reports them.

Forecast-relative monitoring. Monitoring $r_t/\hat{\sigma}_t$ with a fitted and then frozen GARCH(1,1) forecast brought the one-day false alarm rate down to 3.0% to 6.2%, the level of the same monitor with the true conditional volatility (4.7% to 5.3%). Both runs use a plug-in baseline from one day, which the theorems do not cover: with the true baseline the rate for i.i.d. returns is 1.1% to 1.4% (E9), so the plug-in roughly quadruples it, and a level near 5% reflects the slack of Ville’s inequality, not a guarantee. A proof needs the true or a confidence-sequence baseline and a correct forecast model. The price is the signal cap of Proposition 10: at persistence 0.99 a doubling of the long-run variance was detected within one day on only 34% to 43% of paths, with median delays of 1523 to 1791 steps (E10).

Price-level microstructure noise. Noise on log prices makes squared returns positively autocorrelated, so the null fails without a change in volatility: at one-second sampling the one-day false alarm rate on raw returns rose from 5.0% without noise to 41.5% at a noise-to-signal ratio of 4. Over non-overlapping blocks of prices the null holds again for the pre-averaged statistic, so Theorems A and B apply exactly for every block length (ALT1). Supplement S3 gives the details of all three cases.

5.5 Extensions: a running truncation level, many assets and several sampling frequencies

Three extensions are studied in E14, E15 and E17; Supplement S4 reports them in full. First, E14 makes the truncation level a running, data-driven τ_t with a proof, either through a band for the truncated mean that holds for all truncation levels at once (bound $\delta + m/A$) or through rank-based (conformal) bets that need no baseline (bound m/A under exchangeable returns). In simulation the rank detector was valid at its nominal level and two to three times faster than the plug-in for large decreases, while the plug-in with 390 calibration steps alarmed on 40.6% to 42.9% of null paths within five days, where a known baseline would guarantee at most 25% (E14, Q14a and Q14b) [SYNTHETIC]; at a fixed calibration length no bound of the form $C\,m/A$ holds for it (Section 3.4). Second, E15 monitors many assets at once with the e-BH procedure (Wang and Ramdas 2022) applied to mixture e-processes, which controls the false discovery rate within a block under any dependence between assets; in simulation it kept the probability of any false alarm on quiet blocks below 1% with up to 50 assets, where the uncorrected rule reached 37.2%, and found 5.0 percentage points more changes than Bonferroni on average (E15) [SYNTHETIC].

E14 also tests restarts that bet from the first return after an alarm, so that the monitor is never blind. Without a change both versions kept their bounds from the first step after a restart (rank version: 2.7% to 3.4% per 390-step day against 5%, and 15.7% to 17.0% over five days against 25%; band version: at most 0.03%), and the rank version found a change 120 steps after a restart in a median of 20 to 115 steps on 65% to 99% of paths, where the standard protocol, still calibrating, found at most 61% within 2000 steps (E14, Q14f) [SYNTHETIC]. On the five coins at band 0.25 (for the rank version, each payoff divided by 1.25) the restarted monitors watched every minute, against 30% to 33% for the standard protocol, raised about as many alarms (0.66 to 0.88 per day against 0.66 to 0.70) and were confirmed by the next day's volatility about as often (E14, Q14d) [REAL]. The rank version tests exchangeability of the returns after each restart, given the returns up to it, which volatility clustering violates: without a band it alarmed 3.6 to 4.3 times per crypto day. In 30-day synthetic streams in which volatility moves about once a day, the rank restart detected 99.6% (Gaussian) and 98.0% (Student t_3) of the changes within a day, with median delays of 34 and 60 steps, while the monitors that recalibrate for a day after each alarm were blind 57% to 71% of the time and detected 56% to 66% (65% to 66% for the default plug-in) (E14, Q14h) [SYNTHETIC]. In the event design of Section 6.3, the restarted monitors were watching at all 114 releases, against 55 for the standard monitor, and the BTC CPI test, not significant there, gave $p = 0.003$ and 0.0003 (E14, Q14g) [REAL]. Every monitor of E14 costs well under a millisecond per return (Q14e).

Third, E17 runs the detector at 1, 5 and 15 minutes at once, each on non-overlapping blocks with its own truncation level and baseline, and combines the three with weights w_k that sum to 1, because some changes are easier to see in coarser returns. An average of e-values is an e-value (Vovk and Wang 2021), but the usual proof does not cover the plain average checked every minute, because a coarse detector is not a fair bet on the one-minute clock. A union bound proves only three times the budget for it with weights proportional to k. A certified average, which credits each coarse detector between its block ends with the smallest value it can take at the next one, has $\mathbb{P}(\text{false alarm within } m \text{ minutes}) \le \sum_k w_k \lceil m/k \rceil / A$ when each frequency's null holds with the true baselines, for example for i.i.d. returns with any tails; there is no proof with plug-in baselines. At a proved 5% per day and with i.i.d. simulated returns, no average beat the best single frequency, but on persistent changes the equal-weight average was 0% to 5% slower than the one-minute detector and the average with weights proportional to k 11%

to 18% slower. After the lag-one autocorrelation of returns jumped from 0 to 0.3, which the one-minute detector caught on 3.8% of Gaussian paths, the averages of all three frequencies caught 95.5% to 97.3% (83.9% to 89.0% for Student t_3). Under price noise the only proved average, of 5- and 15-minute blocks that share no noisy price, helped only for persistent increases at the strongest noise level, which was added after the first results (E17) [SYNTHETIC]. On the five coins the averages behaved like the one-minute monitor, and at matched alarm counts, in a comparison added after review, the 5- and 15-minute detectors alone were no more precise than it (E17, Supplement S4.3) [REAL].

5.6 Comparison with published change-point methods

E13 compares the e-detector with two published methods [SYNTHETIC]. The first is a nonparametric test for volatility jumps in the spirit of Bibinger, Jirak and Vetter (2017), BJV below: it compares the realised variance of the k returns before each candidate time with that of the k returns after it, standardises the difference and takes the maximum; we use a jump-robust version that drops returns above 4 times a local bipower scale, offline on a fixed window and online as a rolling test with $k = 120$ (and, in a second run on clean Gaussian data, $k = 250$ and 500). This implementation is our reconstruction from the published idea and was not checked against the paper's equations, whose full text we could not reach; every conclusion below therefore uses critical values calibrated by simulation, not the asymptotic Gumbel constants. The second is Bayesian online change-point detection (BOCPD; Adams and MacKay 2007) with Gaussian returns inside a regime and a conjugate prior on the variance, run on raw returns and, as our modification, on returns clipped at the e-detector's level τ. Online, the competitors' thresholds were matched to the e-detector's false alarm rate per 390-step day on clean Gaussian paths (E13).

- *Robustness.* At these matched thresholds BOCPD raised a false alarm within one day on 73.2% of paths with jumps and 53.2% with Student t_5 returns, against 1.3% and 1.2% for the e-detector with a plug-in baseline (measured; this version has no proved bound) and 0.1% and 0.0% with the confidence-sequence baseline, whose 5% bound is proved. Clipping the returns first fixed jumps (1.6%) but not fat tails (7.5% per day under t_5), and the rolling test reached 14.9% under t_5 (E13, Q13b).
- *Speed on clean data.* BOCPD was faster on clean Gaussian data for large decreases (median 26 against 81 steps for a halving of the standard deviation, 51 against 109 for a fall to 0.67 of it) and for a doubling (13 against 20 steps); for a fall to 0.8 of the standard deviation, and with ticks or price noise, it was only 1.0 to 1.9 times faster (137 against 169 steps for sd × 0.8 on Gaussian data, 132 against 132 with ticks) and detected fewer of the sd × 0.8 changes (64.5% to 86.5% against 97.5% to 98.5%). BOCPD on clipped returns kept most of its speed for decreases under jumps (25 against 81 steps for a halving). Neither BOCPD version detected 25% increases as reliably (81% to 90% of them with Gaussian returns or jumps and 46.5% to 84.5% with ticks or price noise, against 97% to 99%) (E13, Q13c).
- *Offline power.* On windows of 2000 steps with critical values matched to 5% size on clean Gaussian windows, the test of the BJV type found a 25% increase of the standard deviation in 6% to 30% of windows with window lengths $k = 50$ or 125 (with or without studentisation), against 100% for the e-detector, which uses the first 500 steps as a change-free calibration stretch. The gap is window length, not betting: with $k = 250$ the same truncated statistic found 76% of 25% increases and 77% of 20% decreases, and with $k = 500$, as long as the e-detector's calibration

stretch, 99% and 100% (E13, Q13a). Long windows have costs: the statistic needs k returns on both sides of the change, it located small changes less precisely (within 25 steps in 49% to 54% of rejections with $k = 500$ against 63% to 65% with $k = 125$), and run online with $k = 500$ it detected 96% of these changes but with median delays of 277 and 326.5 steps, against 104 and 169 for the e-detector (E13, Q13c). The published test is built to find local volatility jumps inside smoothly moving volatility, with short windows and no reference period, so this is not a weakness of the test for its own purpose; the e-detector's advantages are that it needs no window and no change location, and that it runs online.

- *The plug-in caveat again.* In the same offline design, with only 500 calibration steps, the plug-in e-detector's size at its nominal level was 11.4% to 18.6% instead of 5%, while the confidence-sequence baseline gave 0.0% for every law, at a cost of 1.25 to 2.6 times the median delay in the online design (3.7 to 3.9 for decreases with ticks) (E13, Q13a and Q13c).

The e-detector was also the cheapest of the robust online methods, at about 11 microseconds per step against about 300 for BOCPD (E13, Q13e). Section 6.1 compares the methods on real data at equal alarm counts.

6 Real data

All results in this section are **[REAL]**. Real data cannot test the false alarm guarantee, which concerns periods in which volatility does not change, because such periods cannot be identified. What real data can show is how often the monitor alarms, whether its alarms are followed by the volatility they announce, and whether they carry information beyond a forecast. In all runs the settings were fixed before the data were examined: they were frozen after one earlier run on Bitstamp BTC/USD for July to December 2025 (E5), or pre-registered (E7). Returns are divided by the predictable intraday profile of Section 2, the truncation is $c = 3$ scaled MADs of the calibration window, the baseline is the plug-in mean over a calibration window of one day (two sessions for the Nifty 50), missing returns are skipped, and the monitor recalibrates after every alarm. The total alarm level is $A = 28,800$ for a 1440-minute crypto day (each side at $2A$, Proposition 1) and $A = 7500$ for a 375-bar Nifty session, so that with a known baseline the probability of any false alarm is at most 5% per day or session. Because the baseline is a plug-in, this bound is not proved for these runs, and Section 3.4 shows that it is exceeded over horizons of several days. The validation measure is a robust volatility, 1.4826 times the median of $|u|$ over the next day or session, where u is the deseasonalised return; it is not a quantity a trader can trade.

6.1 Five cryptocurrencies

Data. Binance spot one-minute bars for BTC, ETH, XRP, SOL and DOGE against USDT, the coins of Bapat (2026), from a public mirror of the exchange's files with no missing minutes, for 1 July to 31 December 2025 (264,959 returns per coin) and, out of sample, 21 January 2024 to 30 June 2025 (527 days); Bitstamp BTC/USD for a cross-venue check (E6).

Alarms are followed by the announced volatility. With band $b = 0.25$ the monitor raised 0.66 to 0.71 alarms per day for every coin in both periods. In both periods, 75% to 84% of increase alarms and 83% to 93% of decrease alarms were followed by a day whose

robust volatility was on the flagged side of the calibration baseline, against about 50% at random times whose baseline window ended as long before as the alarm's (E6, Q6d). Out of sample these shares fell by only 1 to 4 points. Within 60 minutes of a Binance BTC alarm, Bitstamp had a same-direction alarm 72% of the time, and 73% of Bitstamp alarms had one on Binance, against about 3% for randomly shifted alarm times; across coins about a quarter of alarms coincided with a BTC alarm.

Alarms carry information beyond a forecast, but a tuned ratio rule does as well. At each alarm we forecast the robust volatility of the next 1440 minutes with an EWMA (half-life one day) and with a HAR regression (Corsi 2009) refitted daily, using data up to the alarm only. A rate-matched competitor alarms when the robust volatility of the last S minutes ($S = 60$ or 240) divided by that of the preceding day crosses thresholds that were chosen on the same data to give the same numbers of up and down alarms as the e-detector, with the same one-day dead time after each alarm. This hindsight favours the competitor. Table 7 pools the five coins.

Rule	Period	Up alarms: next day above forecast	Down alarms: next day below forecast
Random times (EWMA / HAR)	Jul to Dec 2025	49% / 51%	51% / 49%
Robust e-detector (EWMA)	Jul to Dec 2025	79% (74 to 82), $n = 367$	81% (75 to 85), $n = 256$
Robust e-detector (HAR)	Jul to Dec 2025	81% (76 to 84)	78% (73 to 83)
Ratio rule, 60 / 240 min (EWMA)	Jul to Dec 2025	74% / 74%	78% / 82%
Random times (EWMA / HAR)	out of sample	50% / 48%	50% / 52%
Robust e-detector (EWMA)	out of sample	77% (74 to 80), $n = 951$	71% (68 to 74), $n = 870$
Robust e-detector (HAR)	out of sample	76% (73 to 79)	74% (71 to 77)
Ratio rule, 60 / 240 min (EWMA)	out of sample	77% / 76%	67% / 71%

Table 7. Share of alarms after which the next day's robust volatility was on the flagged side of a forecast made at the alarm, pooled over five coins, band 0.25, Wilson 95% intervals in brackets (E6, Q6e) [REAL]. Alarms across coins are correlated, so pooled intervals are optimistic.

After an increase alarm the next day's volatility exceeded both forecasts 76% to 81% of the time, against 48% to 51% at random times, and after a decrease alarm it fell short 71% to 81% of the time. The rate-matched ratio rule was within 0 to 5 points at band 0.25, and none of these differences was significant. Only at the strictest band (0.5), out of sample and against the EWMA forecast, was the e-detector better, by about 4 points for increases ($p = 0.03$ to 0.06) and 10 points for decreases against the 60-minute rule ($p < 0.001$); against the HAR forecast the differences were smaller and not significant. The increase alarms of the Gaussian likelihood-ratio monitor were 7 to 14 points less informative than those of the robust monitor. On real data, then, the e-detector does not buy precision over a simple rule tuned with hindsight. What it adds is settings fixed in advance and a false alarm bound that is proved with a known or confidence-sequence baseline; the plug-in runs reported here are not covered by that proof, and on most real days the null is false anyway (Section 6.5).

Against published methods. E13 repeats this check for the competitors of Section 5.6, run continuously with predictable priors and scales and with the same one-day pause after every alarm, and with their up and down thresholds set per series so that each raises as many up and down alarms as the e-detector at band 0.25 (about 0.68 alarms per day). Table 8 pools the five coins.

Method (window in minutes)	Jul to Dec 2025: up / down	Jan to Jun 2025 (held out): up / down
Robust e-detector (one-day baseline)	78.5% / 80.5% (n = 369 / 258)	82.0% / 69.7% (n = 308 / 304)
BOCPD (w = 250)	67.8% / 61.7%	69.2% / 56.7%
BOCPD (w = 720)	70.5% / 66.5%	71.9% / 59.5%
BOCPD (w = 1440)	68.4% / 69.9%	74.2% / 63.9%
BOCPD on truncated returns (w = 250)	68.0% / 71.7%	71.5% / 67.0%
BOCPD on truncated returns (w = 720)	72.6% / 77.9%	74.8% / 69.3%
BOCPD on truncated returns (w = 1440)	69.6% / 81.7%	73.7% / 71.9%
Rolling BJV-type test (k = 120)	65.6% / 58.7%	68.6% / 59.8%
Rolling BJV-type test (k = 480)	69.9% / 74.0%	75.7% / 68.3%
Rolling BJV-type test (k = 1440)	67.9% / 77.7%	69.3% / 73.7%
Rolling BJV-type test, studentised (k = 120)	66.5% / 62.4%	67.8% / 58.3%
Rolling BJV-type test, studentised (k = 480)	68.2% / 71.3%	75.5% / 66.8%
Rolling BJV-type test, studentised (k = 1440)	66.3% / 77.8%	69.3% / 72.7%

Table 8. Share of up (down) alarms after which the next day's robust volatility was above (below) an EWMA forecast made at the alarm, at equal numbers of up and down alarms, pooled over five coins (E13, Q13d) [REAL]. w is the short-run window of BOCPD and k the window of the BJV-type test. At random times the next day is above the forecast 46.5% to 50.1% of the time and below it 49.9% to 53.5% per coin. Rows with windows of 480 to 1440 minutes were added after the first run, with lengths fixed before they were run.

At equal alarm counts and with the first-run windows (2 to 4 hours), the e-detector's alarms were followed by the announced volatility more often than every competitor's (two-proportion tests, $p \le 0.02$; with standard errors clustered by calendar week, which allow for dependence between alarms of the same week on any coin and of either method, $p \le 0.010$), with one exception: in the held-out half-year the down alarms of BOCPD on truncated returns were not significantly different (67.0% against 69.7%, $p = 0.47$); on DOGE the two had the same count (41 of 58 each). The outcome, however, asks about the next day, the time scale of the e-detector's one-day baseline. With windows of a day (BOCPD on truncated returns with w = 720 or 1440, and both BJV-type tests with k = 1440), the competitors' down alarms were as informative as the e-detector's in both half-years (77.7% to 81.7% against 80.5%, and 69.3% to 73.7% against 69.7%; $p \ge 0.28$), while plain BOCPD stayed lower (59.5% to 69.9%). No competitor reached the e-detector's up alarms: the best longer-window competitor, picked after seeing the results, was about 6 points lower (72.6% against 78.5%, $p = 0.065$, week-clustered 0.047; 75.7% against 82.0%,

$p = 0.054$ and 0.028), a borderline difference that is not significant when the two methods are treated as independent samples ($p = 0.21$ and 0.19). Scored on the next four hours instead of the next day, the short-window competitors did as well as the e-detector or better. So the e-detector is not more precise on crypto in general; its alarms are better aligned with day-long shifts than those of competitors with short windows. Rate matching uses the real data, but not the outcome. The e-detector's settings were frozen in E5 on Bitstamp BTC/USD for July to December 2025, so only January to June 2025 is fully held out for it. On the Nifty 50 (2024 and 2025, 494 sessions) all five methods scored 69% to 83% with the first-run windows and no difference was significant; a session-long BJV window did worse (E13, Q13d) [REAL].

6.2 The Nifty 50 index and India VIX

On Nifty 50 one-minute bars from April 2017 to October 2026 (2,361 sessions after removing frozen stretches), with a configuration pre-registered on 2017 to 2021 and a held-out period from 2022 to October 2026, the monitor raised 0.26 alarms per session in both periods. In the held-out period 78% of increase alarms were followed by a session more volatile than the baseline (47% at random times) and 76% of decrease alarms by a calmer one (53%), against 61% for the increase alarms of the Gaussian likelihood-ratio monitor. India VIX agreed with the alarms on the same day but did not move in their direction on the next day more often than on other days. 9 of 11 Union Budget sessions had an alarm, against 25% of ordinary sessions (Fisher $p = 0.0001$), while Reserve Bank of India policy days did not differ. In March 2020 only 6 of 21 sessions had an alarm although volatility was 2.7 times that of the previous 20 sessions: after each alarm the monitor recalibrates on the new level, so it answers "has volatility changed from the recent baseline?", not "is volatility high?" (E7). Details are in Supplement S5.

6.3 FOMC statements and CPI releases

For BTC, ETH and SOL from 21 January 2024 to 31 December 2025, with 16 FOMC statements, 22 CPI releases and same-weekday, same-clock-time control windows, the pre-specified primary test (BTC, the monitor as a user would have run it, Holm correction) found an increase alarm within six hours after 9 of 16 FOMC statements against 14% of weekday-matched controls ($p = 0.0002$); the CPI test was not significant on BTC ($p = 0.11$) but replicated on ETH and SOL. The streaming monitor was recalibrating about 70% of the time; at the FOMC statements at which it was monitoring, it alarmed every time. With a fresh calibration before every anchor (a secondary design added after the first results), 14 to 15 of 16 FOMC statements raised an alarm against 29% to 32% of controls, with a median delay of 23 to 34 minutes. The Gaussian likelihood-ratio monitor also alarmed at almost every release, but within a median of 0 to 2 minutes and on one return of 8 to 9 robust scale units: one large minute is enough to trigger it. It also alarmed in 62% to 76% of control windows, but it ran at band 0 and the robust monitor at band 0.25, so that contrast is partly the band: run at band 0 as well, the robust monitor alarmed in 39% to 51% of control windows (E12). Restarts without a blind period (Section 5.5) were watching at all 114 releases of the three coins, against 55 for the streaming monitor; up alarms within six hours rose from 21 to 24 (band version) and 31 (rank version) of 66 CPI releases and from 26 to 30 and 35 of 48 FOMC statements, at the price of somewhat higher control rates for the rank version, and the BTC CPI cell gave $p = 0.003$ and 0.0003 instead of 0.11. Compared at the same releases, the restarted monitors flagged fewer: at the 26 FOMC statements at which the streaming monitor was watching it flagged all 26,

the band version 19 and the rank version 23; at the 29 such CPI releases it flagged 19, the band version 10 and the rank version 14 (E14, Q14g). Details are in Supplement S5.

6.4 A panel of assets

Run on the five coins with one block per UTC day, the panel monitor of Section 5.5 discovered 86% to 87% of coin-days at band 0.25, so the strict daily null is false on most coin-days and the false discovery guarantee is nearly empty in practice. Compared with Bonferroni, e-BH changed the timing more than the set of discoveries, flagging first on 71% to 77% of the coin-days found by both, and its discoveries pointed the right way (increase discoveries followed by higher volatility 78% to 83% of the time, against 46% to 50% at matched random times) (E15) [REAL]. Details, including the Nifty 50 on its own clock, are in Supplement S5.

6.5 What the real data say about the method

Three lessons qualify the simulation results. First, alarm rates on real data are far above the 5% budget: 0.66 to 0.71 per day on crypto at band 0.25, and 0.14 to 0.57 per session on the Nifty 50 across configurations. This does not contradict the guarantee, which concerns periods in which the null holds and needs a known or confidence-sequence baseline; one-minute volatility relative to a one-day baseline simply moves often. Dividing returns by an EWMA forecast with a half-life of 6 or 24 hours changed the crypto alarm rate by between +2% and −17% (E10, Q10c). Second, a monitor that recalibrates for a full day after every alarm is blind most of the time when alarms come every 1.4 days (about 70% in E12, and 85% with band 0 on Bitstamp BTC/USD, E8). Restarts that bet from the first return after an alarm remove the blind period with about the same alarm and confirmation rates (E14, Section 5.5). Third, on Bitstamp BTC/USD the confidence-sequence baseline of Theorem D reduced the number of alarms by 16% to 20% at every band, almost all of it in decrease alarms, and behaved like a data-driven band of about 0.16 to 0.18 (E8, Q8e). Its value on real data is the proof, not better alarms. Finally, our crypto and Nifty data come from public mirrors of exchange data rather than from the exchanges, and Bapat (2026) does not name the source of its crypto data, so the comparison with its tables in Section 7 uses a data set that is close to, but not the same as, the published one.

7 What a reproduction found about the estimator

Before building the monitor we reimplemented the estimators of Bapat (2026) as printed and compared them with the published simulation and real-data tables. We report four findings, as questions for the author rather than corrections. Where we say what the code behind the tables probably does, this is an inference from published numbers, not a fact. None of the findings touches the central idea, truncation at a robust scale followed by kernel smoothing, and the monitor of this paper uses the scaled MAD and the squared statistic, so none of them affects Sections 3 to 6. We do not claim to have improved the estimator.

Question 1: are the kernel weights normalised? The local estimator $\hat{Q}_t = \frac{1}{3}\sum_{i=t-h}^{t} K((i-t)/h)\, r_i^4$ (eqs. 6 and 12) uses raw kernel weights and no factor for the step length Δ. Under locally constant volatility $\mathbb{E}[\hat{Q}_t] = W(h)\,\sigma_t^4\,\Delta^2$ with $W(h) = \sum_{k=0}^{h} K(k/h)$, about $0.341h + 0.32$ for the Gaussian kernel ($W(30) = 10.6$), so the estimate grows with the

bandwidth. The real-data tables give a clean check: if the kernel estimate is the average of $\hat{Q}_t$ over the sample and RQ is $\frac{1}{3}\sum r_i^4$, their ratio is close to $W(30)/(n-30) \approx 3.99 \times 10^{-5}$ whatever the data. The published ratios are 4.00 to 4.03×10^{-5} for all five coins [PUBLISHED], and our Binance data give 3.986×10^{-5} for all five [REAL] (E1, E6). Once both kernels are put on the same scale, the comparison is fair and the robust idea holds up: with the scaled MAD the normalised robust kernel has a smaller RMSE than the normalised plain kernel in all 45 cells of the simulation design, and with the raw MAD it wins whenever jumps are moderate or large (E1) [SYNTHETIC].

Question 2: is the MAD scaled by 1.4826? Because the MAD shrinks at the same rate as the returns, $\tau/(\sigma\sqrt{\Delta})$ does not change with n, the share of clipped returns stays fixed, and the truncated estimator converges to $\kappa_4(a)$ times integrated quarticity rather than to integrated quarticity: $\kappa_4 = 0.705$ with the raw MAD ($c = 3$ clips at 2.02 standard deviations) and 0.964 with the scaled MAD. In simulation the ratio to the truth was 0.705 at $n = 390$, 1440 and 23,400 (E2) [SYNTHETIC]. Thresholds of the form $c\Delta^{\varpi}$ with $\varpi < 1/2$ (Mancini 2009) avoid this, as does division by κ_4. Two independent checks point to the raw MAD in the published tables: the simulated robust RMSE matches the published one with a median ratio of 0.90 for the raw MAD against 1.29 for the scaled MAD (E1), and on four coins, after adjusting for differences in quartiles between the data sets, the robust kernel matches the published values with ratios of 0.81 to 0.96 for the raw MAD against 1.86 to 2.48 for the scaled MAD (E6) [PUBLISHED against SYNTHETIC and REAL]. For a monitor this constant is a matter of definition, since the null concerns the truncated variance itself; for an estimator of integrated quarticity it has to be divided out.

Question 3: is the averaged estimator meant to equal truncated realised quarticity? Exchanging the two sums in eq. (16), every interior return collects each kernel weight exactly once, so apart from the first and last h returns the averaged estimator equals $W(h)/(3(n-h))\sum_i \tilde{r}_i^4$, a constant times the realised quarticity of the truncated returns. In simulation the normalised averaged robust kernel and plain truncated realised quarticity had correlation 0.98 to 0.99 and mean ratio within 0.4% of 1 in seven scenarios (E2) [SYNTHETIC]. For integrated quarticity, therefore, the kernel and the bandwidth mostly change a constant and the edges. The kernel's real contribution is the local estimate $\hat{Q}_t$, which is what an online procedure needs.

Question 4: is one scale per day or per sample enough when volatility moves within the day? Under stochastic volatility with an intraday pattern, a single daily MAD left a bias of about -11% at $n = 23,400$ after dividing by κ_4 (-14% without), and the corrected estimator's RMSE fell like $n^{-0.14}$, while estimators truncating against a local scale (a few dozen neighbouring returns) converged at the usual $\sqrt{n}$ rate (fitted slopes -0.48 to -0.68 across designs) (ALT3) [SYNTHETIC]. So the first two questions have simple answers under constant volatility (normalise the weights; divide by κ_4), but when volatility moves within the day the truncation also needs a local scale (E2b, ALT3). On one-minute crypto data, 10 to 15 times more returns are clipped within a day than for Gaussian returns, and with one MAD over six months, as in the published crypto table, the robust kernel keeps under 2% of the fourth moment (E6) [REAL]. Large one-minute returns come in bursts (44% to 65% have a large neighbour), and on BTC and DOGE the estimators with one daily MAD threshold come out at 0.33 to 0.46 times the median realised quarticity of Andersen, Dobrev and Schaumburg (2012) (the normalised kernel alone at 0.33 to 0.42), and local plug-ins at 0.66 to 0.73 (ALT3) [REAL]. Was the MAD in the crypto application computed per day, over the whole sample, or locally?

The accompanying documentation lists smaller points in the same spirit (the weight order in eq. 15, the noise model, the constants of the bipower and tripower quarticities, how the RMSE of the real-data table was computed, and the DOGE row, whose zero quartiles imply a zero MAD and hence a zero robust kernel, while a positive value is printed).

8 Discussion, limitations and open problems

What is new and what is not. The probabilistic tools (Ville's inequality, test supermartingales, Shiryaev-Roberts e-detectors, empirical Bernstein confidence sequences, the Bahadur-Savage phenomenon, Lorden's and Lai's lower bounds) are known, and Theorems A and B are instances of known results. What this paper adds is their combination for jump-robust online volatility monitoring: the choice of the clipped squared return with a predictable robust truncation level as the monitored statistic, the conditional-mean nulls and their volatility meaning, the explanation of why a decrease detector needs a bounded statistic and what speed limit truncation then imposes, the confidence-sequence baseline with its proof, the decomposition of the delay cost of robustness, the signal cap under forecast-relative monitoring, and the evidence of Sections 5 and 6. Sections 3.3 and 4 are where we would most value a check by a specialist in sequential analysis.

Limitations.

- *The null is a statement about conditional means.* Persistent volatility clustering, price-level microstructure noise, estimated baselines and errors in the intraday profile all move the conditional mean of X_t without a structural change. Each has a remedy (forecast-relative monitoring, non-overlapping pre-averaged blocks, the confidence-sequence baseline), and each remedy costs speed. The effect of an estimated intraday profile was not isolated in any experiment.
- *Tail shape.* By Lemma 1, a change in the shape of the return distribution at constant volatility, or a leverage effect, can move the truncated variance and raise an alarm that is not a volatility change.
- *Estimated baselines.* The plug-in baseline that the real-data runs use has no proof and exceeds the bound over horizons of several days (Section 3.4). The confidence-sequence baseline has a proof but needs long calibration to be fast (Table 6).
- *Conservatism.* With the default grid the bound is about four times larger than the observed rate at one day (Section 5.2).
- *Speed and precision against published methods.* On Gaussian returns, Bayesian online change-point detection found large decreases two to three times faster at the same false alarm rate (smaller decreases, or decreases with ticks or price noise, 1.0 to 1.9 times faster), and on clipped returns it kept most of that speed under jumps (Section 5.6). On real data, competitors with day-long windows matched the e-detector's down alarms (Section 6.1). The case for the e-detector is robustness, settings fixed in advance with no window to choose, and a guarantee that is proved with a known or confidence-sequence baseline, not speed or precision.
- *Restarts and blindness.* Corollaries C and F bound the expected number of false alarms when the monitor recalibrates after every alarm; they are not a single anytime-valid statement for a whole series, and on real data the standard monitor is blind most of the time (Section 6.5). The restarts of E14 remove the blind period, the faster rank version under the null that the returns after each restart are

exchangeable given the returns up to it, which clustering violates, the band version at a large cost in delay early in a segment (Supplement S4.1).

- *Several assets.* Five coins monitored at 5% per day each give, by the union bound, up to 25% per day for at least one false alarm in the group. Sticky e-BH on mixture e-processes controls the false discovery rate within a block (Section 5.5), but not the probability of at least one false discovery once some assets change, not the false discoveries pooled over many blocks, and on real data the daily null is false on most coin-days (Section 6.4).
- *Delay theory.* The delay results assume i.i.d. post-change data and a known baseline; the second-order formula is a heuristic whose regularity conditions we did not check; Theorem 9 is asymptotic and the asymptotic regime arrives slowly (Section 4.3).
- *Data.* One-minute bars only, from public mirrors of exchange data; no tick data; real-data validation has no ground truth and uses indirect checks (later volatility, forecasts, India VIX, scheduled events).

Open problems.

1. *A certified sharper threshold for mixtures.* For single bets the bound m/A is nearly attained by an adversarial null law, but for the default mixture the worst law found reaches only 42% of it (Section 5.2). A sharper bound that is still provable would translate directly into speed.
2. *A running truncation level and a restart without a blind period, with a proof that costs little.* Real data want τ_t to follow the current scale. E14 gives proved versions of both: a band that holds for all truncation levels at once, rank-based (conformal) bets that need no baseline, and restarts that bet from the first return after an alarm (Section 5.5). The band is slow with short calibration and early in a segment, and the rank bets replace the conditional-mean null by exchangeability. A construction as fast as the plug-in under the conditional-mean null is open.
3. *A narrower baseline.* A betting confidence sequence could recover roughly the gap between our upper bound and the best-constant-bet benchmark of Section 5.3, 7.6 to 10.6 points at 390 calibration steps.
4. *Theory under clustering.* A proof of the cap for the mean of the ratio $\mathbb{E}[r_t^2/s_t]$ that the monitor actually sees, delay bounds for forecast-relative monitoring with estimated forecast parameters, and a principled choice of band for clustered data.
5. *What should the monitor watch?* The same detectors applied to the clipped fourth power were faster for increases and slower for decreases (ALT1). Whether to monitor the truncated variance, the truncated quarticity or both, and how to combine them with one false alarm budget, is a question on which the author of the quarticity estimator is better placed than we are.
6. *Estimation.* A central limit theorem for truncation against a local scale, which the evidence of Section 7 (Question 4) suggests is the right object for one-minute data, and a comparison of classical purely sequential fixed-accuracy estimation of volatility, with a robust quarticity estimate as the variance plug-in, against anytime-valid confidence sequences.

References

Adams, R. P. and MacKay, D. J. C. (2007). Bayesian online changepoint detection. arXiv:0710.3742.

Andersen, T. G. and Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance* 4(2-3), 115-158. doi:10.1016/S0927-5398(97)00004-2

Andersen, T. G., Dobrev, D. and Schaumburg, E. (2012). Jump-robust volatility estimation using nearest neighbor truncation. *Journal of Econometrics* 169(1), 75-93. doi:10.1016/j.jeconom.2012.01.011

Andersen, T. G., Todorov, V. and Zhou, B. (2025). Real-time detection of local no-arbitrage violations. *Quantitative Economics* 16(2), 459-495. doi:10.3982/QE2585

Bahadur, R. R. and Savage, L. J. (1956). The nonexistence of certain statistical procedures in nonparametric problems. *Annals of Mathematical Statistics* 27(4), 1115-1122.

Bapat, S. R. (2026). A jump and noise robust kernel based estimator of quarticity for high frequency data. *International Review of Economics and Finance* 109, 105482. doi:10.1016/j.iref.2026.105482

Barndorff-Nielsen, O. E. and Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics* 2(1), 1-37. doi:10.1093/jjfinec/nbh001

Bibinger, M., Jirak, M. and Vetter, M. (2017). Nonparametric change-point analysis of volatility. *Annals of Statistics* 45(4), 1542-1578. doi:10.1214/16-AOS1499

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* 31(3), 307-327. doi:10.1016/0304-4076(86)90063-1

Boudt, K., Croux, C. and Laurent, S. (2011). Robust estimation of intraweek periodicity in volatility and jump detection. *Journal of Empirical Finance* 18(2), 353-367. doi:10.1016/j.jempfin.2010.11.005

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics* 7(2), 174-196. doi:10.1093/jjfinec/nbp001

Diebold, F. X. (1986). Modeling the persistence of conditional variances: a comment. *Econometric Reviews* 5(1), 51-56.

Dufresne, D. (1990). The distribution of a perpetuity, with applications to risk theory and pension funding. *Scandinavian Actuarial Journal* 1990(1), 39-79.

Fan, X., Grama, I. and Liu, Q. (2015). Exponential inequalities for martingales with applications. *Electronic Journal of Probability* 20, paper 1, 1-22. doi:10.1214/EJP.v20-3496

Grünwald, P., de Heide, R. and Koolen, W. (2024). Safe testing. *Journal of the Royal Statistical Society Series B* 86(5), 1091-1128. doi:10.1093/jrsssb/qkae011

Honda, J. and Takemura, A. (2010). An asymptotically optimal bandit algorithm for bounded support models. *Proceedings of the Conference on Learning Theory (COLT 2010)*.

Horváth, L., Trapani, L. and Wang, S. (2024). Sequential monitoring for explosive volatility regimes. arXiv:2404.17885.

Howard, S. R., Ramdas, A., McAuliffe, J. and Sekhon, J. (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics* 49(2), 1055-1080. doi:10.1214/20-AOS1991

Inclán, C. and Tiao, G. C. (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association* 89(427), 913-923.

Lai, T. L. (1998). Information bounds and quick detection of parameter changes in stochastic systems. *IEEE Transactions on Information Theory* 44(7), 2917-2929.

Lamoureux, C. G. and Lastrapes, W. D. (1990). Persistence in variance, structural change, and the GARCH model. *Journal of Business and Economic Statistics* 8(2), 225-234.

Larsson, M., Ramdas, A. and Ruf, J. (2024). The numeraire e-variable and reverse information projection. arXiv:2402.18810.

Lorden, G. (1970). On excess over the boundary. *Annals of Mathematical Statistics* 41(2), 520-527.

Lorden, G. (1971). Procedures for reacting to a change in distribution. *Annals of Mathematical Statistics* 42(6), 1897-1908.

Mancini, C. (2009). Non-parametric threshold estimation for models with stochastic diffusion coefficient and jumps. *Scandinavian Journal of Statistics* 36, 270-296.

Mikosch, T. and Starica, C. (2004). Nonstationarities in financial time series, the long-range dependence, and the IGARCH effects. *Review of Economics and Statistics* 86(1), 378-390. doi:10.1162/003465304323023886

Page, E. S. (1954). Continuous inspection schemes. *Biometrika* 41(1-2), 100-115. doi:10.1093/biomet/41.1-2.100

Pollak, M. (1985). Optimal detection of a change in distribution. *Annals of Statistics* 13(1), 206-227. doi:10.1214/aos/1176346587

Pollak, M. (1987). Average run lengths of an optimal method of detecting a change in distribution. *Annals of Statistics* 15(2), 749-779. doi:10.1214/aos/1176350373

Ram, A. and Ramdas, A. (2026). Asymptotically optimal sequential change detection for bounded means. arXiv:2602.05272.

Ramdas, A., Grünwald, P., Vovk, V. and Shafer, G. (2023). Game-theoretic statistics and safe anytime-valid inference. *Statistical Science* 38(4), 576-601. doi:10.1214/23-STS894

Roberts, S. W. (1966). A comparison of some control chart procedures. *Technometrics* 8(3), 411-430.

Shafer, G. (2021). Testing by betting: a strategy for statistical and scientific communication. *Journal of the Royal Statistical Society Series A* 184(2), 407-431. doi:10.1111/rssa.12647

Shin, J., Ramdas, A. and Rinaldo, A. (2024). E-detectors: a nonparametric framework for sequential change detection. *New England Journal of Statistics in Data Science* 2(2), 229-

260. doi:10.51387/23-NEJSDS51

Shiryaev, A. N. (1963). On optimum methods in quickest detection problems. *Theory of Probability and Its Applications* 8(1), 22-46. doi:10.1137/1108002

Ville, J. (1939). *Étude critique de la notion de collectif*. Thèse, Paris.

Vovk, V., Nouretdinov, I. and Gammerman, A. (2003). Testing exchangeability on-line. *Proceedings of the International Conference on Machine Learning (ICML 2003)*, 768-775.

Vovk, V. and Wang, R. (2021). E-values: calibration, combination and applications. *Annals of Statistics* 49(3), 1736-1754. doi:10.1214/20-AOS2020

Wang, R. and Ramdas, A. (2022). False discovery rate control with e-values. *Journal of the Royal Statistical Society Series B* 84(3), 822-852. doi:10.1111/rssb.12489

Waudby-Smith, I. and Ramdas, A. (2024). Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B* 86(1), 1-27. doi:10.1093/jrsssb/qkad009

Woodroofe, M. (1982). *Nonlinear Renewal Theory in Sequential Analysis*. CBMS-NSF Regional Conference Series in Applied Mathematics 39. SIAM, Philadelphia.

Appendix A. Proofs for the estimated baseline (Section 3.4)

Throughout, the observations of the confidence sequence are indexed by $i = 1, 2, \dots$ (global time $n_\tau + i$), $\mathcal{F}_{i-1}$ is the information before the i-th observation, $Z_i = \min(r_i^2, \tau^2)/\tau^2 \in [0,1]$ and $\eta_i = \mathbb{E}[Z_i \mid \mathcal{F}_{i-1}]$. The one-sided ranges are $a_i^{\uparrow} = 1 - \hat{z}_{i-1}$ and $a_i^{\downarrow} = \hat{z}_{i-1}$, so that $\hat{z}_{i-1} - Z_i \ge -a_i^{\uparrow}$ and $Z_i - \hat{z}_{i-1} \ge -a_i^{\downarrow}$ always hold.

Lemma A.1. Let $a > 0$ and $\lambda \ge 0$ with $\lambda a < 1$, and $\psi_a(\lambda) = (-\log(1-\lambda a) - \lambda a)/a^2 \ge 0$. Then for every $\xi \ge -a$,

$$\log(1+\lambda\xi) \ \le\ \lambda\xi - \psi_a(\lambda)\,\xi^2, \qquad \text{equivalently} \qquad \exp(\lambda\xi - \psi_a(\lambda)\xi^2) \le 1 + \lambda\xi.$$

Proof. For $x > -1$, Taylor's formula with integral remainder gives $\log(1+x) - x = x^2\phi(x)$ with $\phi(x) = -\int_0^1 (1-s)(1+sx)^{-2}\,ds$. For each $s \in [0,1]$ the map $x \mapsto (1+sx)^{-2}$ is nonincreasing on $(-1, \infty)$, so ϕ is nondecreasing. Put $x = \lambda\xi \ge -\lambda a > -1$. Then $\log(1+\lambda\xi) - \lambda\xi \ge (\lambda\xi)^2\phi(-\lambda a)$, and $\phi(-\lambda a) = (\log(1-\lambda a) + \lambda a)/(\lambda a)^2$, so the right side equals $-\psi_a(\lambda)\xi^2$. For $\lambda = 0$ both sides are 0. ▫

With $a = 1$ this is the inequality of Fan, Grama and Liu (2015) behind the empirical Bernstein confidence sequences of Howard et al. (2021) and Waudby-Smith and Ramdas (2024).

Theorem D.1 (time-uniform coverage). For the construction of Section 3.4, without any assumption on the dependence between observations or on how η_i varies,

$$\mathbb{P}(\bar{\eta}_t^{\uparrow} \ge U_t \ \text{for some } t \ge 1) \le \delta_u, \qquad \mathbb{P}(\bar{\eta}_t^{\downarrow} \le L_t \ \text{for some } t \ge 1) \le \delta_l.$$

Proof. Upper bound. Let $K_0 = 1$ and $K_t = \prod_{i \le t} \exp(\lambda_i^{\uparrow}(\eta_i - Z_i) - \psi_i^{\uparrow} e_i)$. Write $\eta_i - Z_i = \xi_i + (\eta_i - \hat{z}_{i-1})$ with $\xi_i = \hat{z}_{i-1} - Z_i \ge -a_i^{\uparrow}$, and note $e_i = \xi_i^2$. Lemma A.1 with $a = a_i^{\uparrow}$ and $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = \hat{z}_{t-1} - \eta_t$ give

$$\mathbb{E}[K_t/K_{t-1} \mid \mathcal{F}_{t-1}] \le \mathbb{E}[1+\lambda_t^{\uparrow}\xi_t \mid \mathcal{F}_{t-1}]\, e^{\lambda_t^{\uparrow}(\eta_t-\hat z_{t-1})} = (1-y)e^{y} \le 1, \qquad y=\lambda_t^{\uparrow}(\eta_t-\hat z_{t-1}),$$

since $(1-y)e^{y} \le 1$ for every real y. So K is a nonnegative supermartingale with $K_0 = 1$, and Ville's inequality gives $\mathbb{P}(K_t \ge 1/\delta_u \text{ for some } t) \le \delta_u$. Taking logarithms, $K_t \ge 1/\delta_u$ is the event $\sum_{i\le t}\lambda_i^{\uparrow}\eta_i \ge \sum_{i\le t}\lambda_i^{\uparrow}Z_i + \sum_{i\le t}\psi_i^{\uparrow}e_i + \log(1/\delta_u)$, that is $\bar\eta_t^{\uparrow} \ge U_t$. Lower bound: the same argument with $K_t' = \prod_{i\le t}\exp(\lambda_i^{\downarrow}(Z_i-\eta_i) - \psi_i^{\downarrow}e_i)$, $\xi_i = Z_i - \hat z_{i-1} \ge -a_i^{\downarrow}$ and $(1+y)e^{-y} \le 1$. ▫

Since $0 \le \bar\eta \le 1$, the clipped bounds $\min(U_t, 1)$ and $\max(L_t, 0)$ fail on the same events. The default predictable choices follow Waudby-Smith and Ramdas (2024):
$\hat z_t = (1/2 + \sum_{i\le t} Z_i)/(t+1)$, a variance estimate $\hat\sigma_t^2 = (1/4 + \sum_{i\le t}(Z_i - \hat z_i)^2)/(t+1)$, and
$\lambda_t = \min(\sqrt{2\log(1/\delta)/(d\,\hat\sigma_{t-1}^2)}, c_0/a_t)$ with $c_0 = 0.75$ and d a planned sample size. Validity does not depend on these choices.

Proof of Theorem D. We write U_{T_0} and L_{T_0} for the bounds computed from the observations up to global time T_0, and similarly $\bar\eta_{T_0}^{\uparrow}$, $\bar\eta_{T_0}^{\downarrow}$. Let $G^{\uparrow} = \{\bar\eta_{T_0}^{\uparrow} < U_{T_0}\}$, $G^{\downarrow} = \{\bar\eta_{T_0}^{\downarrow} > L_{T_0}\}$ and $G = G^{\uparrow} \cap G^{\downarrow}$; these events are $\mathcal{F}_{T_0}$-measurable.

Step 1 (the bounds hold at the random time T_0). Theorem D.1 applies to the observations after n_τ, because τ is fixed before they start. The complement of $G^{\uparrow}$ is contained in the event that the upper bound fails at some time, whatever T_0 is, so $\mathbb{P}((G^{\uparrow})^c) \le \delta_u$, $\mathbb{P}((G^{\downarrow})^c) \le \delta_l$ and $\mathbb{P}(G^c) \le \delta$.

Step 2 (on G both statistics satisfy Lemma 2). Let $t > T_0$. On G, $(H^{\uparrow})$ gives $\mu_t \le (1+b)\tau^2\bar\eta_{T_0}^{\uparrow} \le (1+b)\tau^2\min(U_{T_0}, 1) = m^{\uparrow}$, so $\mathbb{E}[f_t^{\uparrow} \mid \mathcal{F}_{t-1}] = 1 + \lambda(\mu_t/m^{\uparrow} - 1) \le 1$, and $f_t^{\uparrow} \ge 1 - \lambda > 0$. If the decrease side is on, $(H^{\downarrow})$ gives $\mu_t \ge (1-b)\tau^2\bar\eta_{T_0}^{\downarrow} > (1-b)\tau^2\max(L_{T_0}, 0) = m^{\downarrow}$, so $\mathbb{E}[f_t^{\downarrow} \mid \mathcal{F}_{t-1}] \le 1$, and $f_t^{\downarrow} \ge 1 - \lambda(B-1) > 0$ because $X_t \le \tau^2$. The bets are fixed at T_0, hence predictable. As in Lemma 2, $\bar M_t = (M_t^{\uparrow} + M_t^{\downarrow})/2$ satisfies $\mathbb{E}[\bar M_t \mid \mathcal{F}_{t-1}] \le \bar M_{t-1} + 1$ on $G \cap \{t > T_0\}$; if the decrease side is off, $M^{\downarrow} = 0$ and the same holds. A skipped (missing) return leaves $\bar M$ unchanged; when, given the past, whether r_t is missing carries no information about its size, the bound on $\mathbb{E}[f_t \mid \mathcal{F}_{t-1}]$ also holds given that r_t is observed, and the inequality is preserved.

Step 3 (Ville's inequality after a random start). Fix s and let $E_s = G \cap \{T_0 = s\} \in \mathcal{F}_s$. For $s \le t \le s+m$ put $S_t = 1_{E_s}(\bar M_t + m - (t-s))$ with $\bar M_s = 0$. Since $E_s \in \mathcal{F}_{t-1}$, Step 2 gives $\mathbb{E}[S_t \mid \mathcal{F}_{t-1}] \le S_{t-1}$; moreover $S \ge 0$ and $S_s = m\,1_{E_s}$. Ville's inequality gives $\mathbb{P}(\max_{s<t\le s+m} S_t \ge A) \le m\,\mathbb{P}(E_s)/A$. An alarm requires $\max(M^{\uparrow}, M^{\downarrow}) \ge 2A$, hence $\bar M \ge A$ and $S \ge A$. Summing over s, $\mathbb{P}(G \cap \{\text{alarm within } m\}) \le m\,\mathbb{P}(G)/A$.

Step 4 (part 1). $\mathbb{P}(\text{alarm within } m) \le \mathbb{P}(G^c) + \mathbb{P}(G)\,m/A \le \delta + m/A$. When $m < A$ the map $p \mapsto p + (1-p)m/A$ is increasing, which gives $\le \delta + (1-\delta)m/A$.

Part 2. Let ρ be the first $t > T_0$ with $\bar M_t \ge A$; the alarm comes no earlier. For E_s as above and $n \ge 1$, $N = \min(\rho, s+n)$ is a bounded stopping time and $1_{E_s}(\bar M_t - (t-s))$, $t \ge s$, is a supermartingale by Step 2. Optional stopping gives $\mathbb{E}[1_{E_s}(N-s)] \ge \mathbb{E}[1_{E_s}\bar M_N] \ge A\,\mathbb{P}(E_s \cap \{\rho \le s+n\})$, and monotone convergence as $n \to \infty$ gives $\mathbb{E}[1_{E_s}(\rho - s)] \ge A\,\mathbb{P}(E_s)$ (if $\rho = \infty$ with positive probability on E_s the left side is infinite). Summing over s, $\mathbb{E}[\rho - T_0] \ge \mathbb{E}[1_G(\rho - T_0)] \ge A\,\mathbb{P}(G) \ge (1-\delta)A$.

Part 3. The same proof with one statistic at level A and only $G^{\uparrow}$ (or $G^{\downarrow}$). ▫

Proof of Corollary E. Since τ is fixed at time n_τ and the conditional law of r_t is P_0 for $t > n_\tau$ before the change, $\mu_t = \mathbb{E}_{P_0}[\min(r^2, \tau^2)]$ is the same constant μ for all such t, every bet-weighted average of the $\eta_i = \mu/\tau^2$ equals μ/τ^2, and $(H^\uparrow)$, $(H^\downarrow)$ hold with $b = 0$. Apply Theorem D. ▫

Proof of Corollary F. Let G_k be the event that both bounds of segment k hold at its end of calibration, and split $N_m = N_{\text{bad}} + N_{\text{good}}$ according to whether the alarm's segment satisfies G_k.

Bad segments. Segment k starts at a stopping time S_k, and its bounds use only returns after S_k, so Theorem D.1 applied conditionally on $\mathcal{F}_{S_k}$ gives $\mathbb{P}(G_k^c \mid \mathcal{F}_{S_k}) \le \delta$ on $\{S_k < \infty\}$. Let $C_k \in \mathcal{F}_{S_k}$ be the event that fewer than m monitored steps have passed before S_k. A segment raises at most one alarm, and only segments with C_k can contribute to N_m, so $\mathbb{E}[N_{\text{bad}}] \le \sum_k \mathbb{E}[1_{C_k}\,\mathbb{P}(G_k^c \mid \mathcal{F}_{S_k})] \le \delta\,\mathbb{E}[\#\{k : C_k\}]$. Segment $k+1$ starts right after alarm k, so C_{k+1} implies that alarm k is among the first m monitored steps; hence $\#\{k : C_k\} \le 1 + N_m$ and $\mathbb{E}[N_{\text{bad}}] \le \delta(1 + \mathbb{E}[N_m])$.

Good segments. Let W_t be the sum of $\bar{M}$ at all past alarms of good segments, plus the current $\bar{M}$ if the current segment is good and monitoring, minus the number of monitored steps so far in good segments. Whether a segment is good is known at the end of its calibration, so by Step 2 above every monitored step of a good segment has $\mathbb{E}[W_t - W_{t-1} \mid \mathcal{F}_{t-1}] \le 0$; at an alarm the statistic moves into the sum unchanged, and in all other steps W does not move. So W is a supermartingale with $W_0 = 0$. Let σ be the time of the m-th monitored step. Each alarm of a good segment adds at least A, so $W_{\sigma \wedge n} \ge A N_{\text{good}}(\sigma \wedge n) - m$; optional stopping at $\sigma \wedge n$ and monotone convergence give $\mathbb{E}[N_{\text{good}}] \le m/A$.

Adding, $\mathbb{E}[N_m] \le \delta(1 + \mathbb{E}[N_m]) + m/A$; since $N_m \le m$ is finite, this rearranges to $\mathbb{E}[N_m] \le (\delta + m/A)/(1 - \delta)$. ▫

Why the sample split. If τ is computed from the whole calibration window, each X_i in the window depends through τ on later returns, so X_i is not adapted and K in Theorem D.1 need not be a supermartingale. Two other valid routes are a union bound over a grid of truncation levels fixed in advance (costing $\log(K/\delta)$ instead of $\log(1/\delta)$ for K levels) and a band that holds for all truncation levels at once (E14). Because the coverage of Theorem D.1 is uniform in time, the calibration may also stop at a data-dependent time, for example when the relative width $(\min(U_t, 1) - \max(L_t, 0))/\bar{Z}_t$ first falls below a target; a fixed-n interval would not allow this, since a normal-approximation interval checked at every n excludes the true mean at some n with probability one.

Appendix B. Proofs for the detection delay (Section 4)

Notation is that of Section 4.1: $V = v(Y)$ is $Y - 1$ (increase) or $1 - Y$ (decrease), with values in $[v_{\text{lo}}, v_{\text{hi}}]$, $v_{\text{lo}} < 0 < v_{\text{hi}}$, and $\lambda_{\max} = 1/|v_{\text{lo}}|$. Since v is affine and one-to-one, a function of y can be viewed as a function of v.

Proof of Lemma 4. If Q is not absolutely continuous with respect to P, $\mathrm{KL}(Q\|P) = \infty$. Otherwise let $q = dQ/dP$. By Jensen's inequality,
$\mathbb{E}_Q[\log E] - \mathrm{KL}(Q\|P) = \mathbb{E}_Q[\log(E/q)] \le \log \mathbb{E}_Q[E/q] = \log \mathbb{E}_P[E\,1\{q > 0\}] \le \log \mathbb{E}_P[E] \le 0$. ▫

Proof of Lemma 5. (a) Let $E \geq 0$ with $\mathbb{E}_P[E] \leq 1$ for every law P on $[0, B]$ with $\mathbb{E}_P[V] \leq 0$. First, E is bounded: for $v > 0$, the two-point law with mass $p = |v_{\mathrm{lo}}|/(v + |v_{\mathrm{lo}}|)$ at v and $1 - p$ at v_{lo} has mean 0, so $pE(v) \leq 1$ and $E(v) \leq 1 + v/|v_{\mathrm{lo}}|$; for $v \leq 0$ the point mass gives $E(v) \leq 1$. Let φ be the upper concave envelope of E on $[v_{\mathrm{lo}}, v_{\mathrm{hi}}]$. The largest value of $\mathbb{E}_P[E]$ over laws with $\mathbb{E}_P[V] = a$ is $\varphi(a)$ (two-point laws suffice), so $\varphi(a) \leq 1$ for every $a \leq 0$. As φ is concave and finite and 0 is interior, it has a supergradient s at 0: $\varphi(v) \leq \varphi(0) + sv \leq 1 + sv$. If $s \geq 0$, take $\lambda = s$; since $0 \leq E(v_{\mathrm{lo}}) \leq 1 + sv_{\mathrm{lo}}$, $s \leq \lambda_{\max}$. If $s < 0$, then $\varphi(v) \leq \varphi(0) \leq 1$ for $v \geq 0$, and $\varphi \leq 1$ for $v \leq 0$ by assumption; take $\lambda = 0$.

b. "$\leq$": each bet $1 + \lambda v$ is an e-variable for every $P \in \mathcal{H}$, so Lemma 4 gives $g(\lambda) \leq \mathrm{KL}(Q\|P)$; take the infimum over P and the maximum over λ. "$\geq$": at an interior maximiser, $g'(\lambda^*) = \mathbb{E}_Q[V/(1 + \lambda^* V)] = 0$. The measure $P^*(dy) = Q(dy)/(1 + \lambda^* v(y))$ has total mass $\mathbb{E}_Q[1/(1 + \lambda^* V)] = 1 - \lambda^* \mathbb{E}_Q[V/(1 + \lambda^* V)] = 1$ and mean of V equal to $\mathbb{E}_Q[V/(1 + \lambda^* V)] = 0$, so $P^* \in \mathcal{H}$, and $\mathrm{KL}(Q\|P^*) = \mathbb{E}_Q[\log(1 + \lambda^* V)] = g(\lambda^*)$. ▫

Proof of Corollary 6. The first inequality is the data processing inequality, since X is a function of r. The second holds because $P_{0,Y} \in \mathcal{H}$, so the infimum defining $\mathrm{KL}_{\inf}(Q)$ is at most $\mathrm{KL}(Q_Y\|P_{0,Y}) = \mathrm{KL}(Q_X\|P_{0,X})$ (Y is a one-to-one function of X). The third holds because each bet is an e-variable for every $P \in \mathcal{H}$, so Lemma 4 gives $g(\lambda_k) \leq \mathrm{KL}(Q\|P)$ for every such P. ▫

Proof of Corollary 7. Each bet satisfies $1 + \lambda v \leq 1 + \lambda_{\max} v_{\mathrm{hi}}$, which is B on the increase side ($\lambda_{\max} = 1$, $v_{\mathrm{hi}} = B - 1$) and $B/(B - 1)$ on the decrease side ($\lambda_{\max} = 1/(B - 1)$, $v_{\mathrm{hi}} = 1$). By Lemma 5(a) every e-variable for $\mathcal{H}$ obeys the same bound, and so does $\mathrm{KL}_{\inf} = g(\lambda^*)$ when λ^* is interior (Lemma 5(b)). For the detector, if every factor of bet k equals its maximum F_k, then $M_n^{(k)} = \sum_{j \leq n} F_k^{n-j+1}$, and any other data give a smaller value. ▫

Proof of Theorem 9. Write $g^* = g(\lambda^*) = \mathrm{KL}_{\inf}(Q)$. For $0 < b' < \lambda_{\max}$ let $\bar{v} = \max(|v_{\mathrm{lo}}|, v_{\mathrm{hi}})$, $C = \bar{v}^2/(1 - b'|v_{\mathrm{lo}}|)^2$, which bounds $|g''|$ and $|\partial_\lambda^2 \log(1 + \lambda V)|$ on $[0, b']$, and $o = \log(1 + b'v_{\mathrm{hi}})$, which bounds every log factor of a bet $\lambda \leq b'$ and hence every overshoot. By Theorem 8(b) it suffices to take $v = 1$.

a. Let $P_r^*(dr) = Q_r(dr)/(1 + \lambda^* v(r))$, where $v(r) = v(Y(r))$. As in Lemma 5(b) it is a probability law with $\mathbb{E}_{P^*}[V] = 0$, so $\mathbb{E}_{P^*}[\min(r^2, \tau^2)] = m^{\uparrow}$ ($m^{\downarrow}$ on the decrease side) and i.i.d. returns from P_r^* satisfy the null. Since $1 + \lambda^* v$ is bounded away from 0 and ∞, Q_r and P_r^* are mutually absolutely continuous, and $\mathrm{KL}(Q_r\|P_r^*) = \mathbb{E}_Q[\log(1 + \lambda^* V)] = g^*$. A rule with average run length at least A under every i.i.d. null law has it under P_r^*, and Lorden's (1971) lower bound for a change from P_r^* to Q_r gives $\mathrm{WADD}_Q(T) \geq (\log A/g^*)(1 + o(1))$.

b. The upper half is Theorem 8(a) with the best bet. For the lower half, let $S_n^k = \sum_{i \leq n} \log(1 + \lambda_k V_i)$ and $R_N^k = \max_{0 \leq i < n \leq N} (S_n^k - S_i^k)$. Writing $S_n^k = g_k n + D_n^k$ with D^k a zero-mean random walk, $R_N^k \leq \max(g_k, 0)N + 2\max_{j \leq N} |D_j^k| = \max(g_k, 0)N + o(N)$ almost surely by the strong law of large numbers. Since $M_n \leq n \max_{k,j} L_n^{(j,k)}$, for all $n \leq N$ we have $\log M_n \leq \log N + \max_k R_N^k \leq g_{\mathrm{grid}} N + o(N)$ almost surely. With $N_A = (1 - \varepsilon)\log A/g_{\mathrm{grid}}$ this gives $\mathbb{P}(T_A \leq N_A) \to 0$, so $\mathbb{E}[T_A] \geq N_A(1 - o(1))$; let $\varepsilon \to 0$ and divide by $\log A/\mathrm{KL}_{\inf}$.

c. With K equally spaced bets on $[a, b']$ and spacing $h = (b' - a)/(K - 1)$, some bet satisfies $|\lambda_k - \lambda^*| \leq h/2$. Since $g'(\lambda^*) = 0$ and $|g''| \leq C$, Taylor's theorem gives $g(\lambda_k) \geq g^* - Ch^2/8$, and Theorem 8(a) with $w_k = 1/K$ gives $\mathrm{WADD}(T_A) \leq (\log A + \log K + o)/(g^* - Ch^2/8)$. With

$K_A = 1 + \lceil (b' - a)\sqrt{\log A} \rceil$, $h \le 1/\sqrt{\log A}$ and $\log K_A \le \frac{1}{2} \log \log A + \log (2 + b' - a)$ for $\log A \ge 1$, and the denominator is $g^* (1 - O(1/\log A))$, which gives the claim, uniformly over laws with $\lambda^* \in [a, b']$ and g^* bounded away from 0. Balancing $\log K$ against $(\log A) C (b' - a)^2 / (8 g^* K^2)$ shows that $K \asymp \sqrt{\log A}$ minimises the second-order term of this bound.

d. Take $\varepsilon > 0$ with $[\lambda^* - \varepsilon, \lambda^* + \varepsilon] \subset [a, b']$ and write $h_i(\lambda) = \log (1 + \lambda V_i)$. Each h_i is concave, so on the interval it is at least $Z_i = \min (h_i(\lambda^* - \varepsilon), h_i(\lambda^* + \varepsilon))$. Keeping only the start time 1 and the interval, $M_t \ge \int_{\lambda^* - \varepsilon}^{\lambda^* + \varepsilon} \prod_{i \le t} (1 + \lambda V_i) \pi(d\lambda) \ge 2\varepsilon p_0 \exp (\sum_{i \le t} Z_i)$. The Z_i are i.i.d. and bounded, and Taylor's theorem gives $\mathbb{E}[Z] \ge g^* - \varepsilon D - C\varepsilon^2/2$ with $D = \mathbb{E}_Q |V/(1 + \lambda^* V)|$. The Wald argument of Theorem 8(a) for the level $\log (A/(2\varepsilon p_0))$ gives $\mathbb{E}[T_A] \le (\log A + \log (1/\varepsilon) + \log (1/(2p_0)) + o)/(g^* - \varepsilon D - C\varepsilon^2/2)$, and $\varepsilon = 1/\log A$ gives the claim. ▫

A Laplace approximation of the mixture suggests that (d) can be sharpened to $(\log A + \frac{1}{2} \log \log A + O(1))/\mathrm{KL}_{\inf}$, the order of the refined grid; we have not proved this.

Cost of a gap in the grid. If λ^* is not a bet, $\mathrm{KL}_{\inf} - g(\lambda_k) = \frac{1}{2}\sigma_*^2 (\lambda_k - \lambda^*)^2 + O(|\lambda_k - \lambda^*|^3)$ with $\sigma_*^2 = \mathbb{E}_Q [V^2/(1 + \lambda^* V)^2]$. For a small change, with $\mathbb{E}_Q V = d$ and $\mathrm{Var}_Q V = s^2$, $\lambda^* \approx d/s^2$, $\mathrm{KL}_{\inf} \approx d^2/(2s^2)$ and $g(\lambda)/\mathrm{KL}_{\inf} \approx 1 - (\lambda/\lambda^* - 1)^2$: a geometric grid whose neighbouring bets differ by a factor ρ keeps at least $1 - ((\rho - 1)/(\rho + 1))^2$ of the growth between its smallest and largest bets (0.89 for $\rho = 2$). A gap costs $\log A (1/g_{\text{grid}} - 1/\mathrm{KL}_{\inf})$ steps, which grows with A, while an extra bet costs about $\log ((K + 1)/K)$ nats, which does not.

Appendix C. Settings of the experiments

All simulations use fixed seeds, and all settings were fixed before the reported runs. Per-step units: the pre-change diffusive standard deviation is 1 unless stated. "Plug-in" means $\tau = 3$ scaled MADs and m_0 the mean of X over the calibration window; "true baseline" means both computed from the distribution.

Label	Returns	Baseline and calibration	Alarm level	Paths	Change design
E4	Gaussian; jumps (prob. 0.005, sd 8); Student t_5; price noise (sd 0.5 on log prices); slow SV (log-sd 0.15, half-life 390)	plug-in, 2000 steps, same law	$A = 7800$ per side	300 per cell	sd ratio 1.25, 1.5, 2, 0.5, 0.67, 0.8 at monitored step 500
E4e	Gaussian, Student t_3, jumps (0.005, sd 8)	true or plug-in (390, 1440 steps)	$A = 7800$; restarts $A = 300$	4000; restarts 400 (true), 40 (streaming)	none

Label	Returns	Baseline and calibration	Alarm level	Paths	Change design
E8	Gaussian, jumps (0.005, sd 8), Student t_5	plug-in, plug-in plus one standard error, or confidence sequence (n_τ = 25% of 390, 1440, 5000 steps; $\delta = 0.01$ per side)	7800 (plug-in), 9750 (CS)	10,000 null, 2000 change	as E4, at step 500
E9	Gaussian, Student t_3, jumps (0.005, sd 10), ticks of 0.5 sd	true (m_0 from 2×10^7 draws)	$A = 7800$, two-sided	4000	none
E9c	GARCH(1,1), unit unconditional variance, persistence 0.95 to 0.9999	true unconditional	$A = 28,800$ (1440-step day)	2000	none
E10	GARCH(1,1), $(\alpha, \beta) = (0.05, 0.90), (0.05, 0.94), (0.03, 0.969)$	2000 warm-up steps, plug-in on one 1440-step day; GARCH forecast fitted by QML with variance targeting on a separate 30,000-step path	$A = 28,800$	600 (null), 300 per row (breaks); 200 paths of 18,000 steps (cap)	scale or intercept break 200 steps after calibration, 5 days
E11	Gaussian, jumps (0.005, sd 8), Student t	true, $c = 3$ unless varied	$A = 7800$ (and 10^4 to 10^8)	3000 per case	change at the first monitored step
E11f	Gaussian, Student t_4, jumps (0.005, sd 8), ticks	plug-in, 1440 steps	$A = 28,800$	2000 null, 600 per change	sd ratio 0.8, 0.67, 0.5 at step 1

Label	Returns	Baseline and calibration	Alarm level	Paths	Change design
E13	Gaussian; jumps (0.005, sd 8); Student t_5; price noise (sd 0.5); ticks of 0.5 sd; slow SV; GARCH(1,1) with persistence 0.99	offline: 500 change-free steps of a 2000-step window (plug-in or CS); online: plug-in or CS on 2000 steps	offline: critical values matched to 5% on clean Gaussian windows; online: $A = 7800$, competitors matched to the e-detector's rate per 390-step day on clean Gaussian paths	offline 1000 null, 300 per change; online 1000 null, 200 per change	offline at step 1000; online at monitored step 500; sd ratio 1.25, 1.5, 2, 0.8, 0.67, 0.5
E14	Gaussian, Student t_3, jumps (0.005, sd 10), ticks of 0.5 sd	plug-in, running plug-in, CS, band over 13 truncation levels, ranks (390 or 1440 steps); restarts with no calibration	$A = 7800$ (m/A) or 9750 ($\delta + m/A$); streams $A = 28,800$ or 36,000	5000 null, 1000 per change; streams 40 paths of 30 days	at monitored step 500; after a restart at 30, 120 or 360 steps; streams about one change per day
E17	Gaussian, Student t_3, jumps (0.005, sd 8); price noise $u_t \sim N(0, \gamma^2)$ on log prices, γ = 0.5, 1, 2	true (2 million simulated blocks) or plug-in (1440 minutes, all block phases)	proved 5% per 1440-minute day: A = 28,800, 5760, 1920 for 1, 5, 15 minutes alone, 2880 to 12,160 for combinations	10,000 null days, 2000 per change	at monitored minute 247: sd ratio 1.25, 1.5, 2, 0.8, 0.67, 0.5; bursts ×2 or ×3 for 15 or 60 minutes; lag-one autocorrelation from 0 to 0.3; $\gamma = 2$ added after the $\gamma = 1$ results
ALT1 (noise)	one-second returns, 23,400 per day, i.i.d. Gaussian price noise	per experiment	A = steps per day / 0.05	200 days per cell	none (validity)

Label	Data [REAL]	Settings
E6	Binance BTC, ETH, XRP, SOL, DOGE (USDT), 1 July to 31 December 2025 and 21 January 2024 to 30 June 2025; Bitstamp BTC/USD, July to December 2025	$A = 28,800$, $c = 3$, calibration 1440 minutes, bands 0, 0.25, 0.5, 20-day past-only intraday profile, recalibration after each alarm; controls: random times with matched delay, EWMA and HAR forecasts, rate-matched ratio rule

Label	Data [REAL]	Settings
E7	Nifty 50, April 2017 to October 2026, 2,361 sessions	$A = 7500$ per 375-bar session, calibration 375 or 750 bars, bands 0, 0.25, 0.5, session-aware profile; primary 750 bars and band 0.25; development 2017 to 2021, held-out 2022 to October 2026; India VIX; event dates verified individually
E12	Binance BTC, ETH, SOL, 21 January 2024 to 31 December 2025	as E6 with band 0.25; 16 FOMC statements, 22 CPI releases; 134 and 299 control windows, matched by weekday; stratified permutation tests, Holm correction over the primary family
E13	Binance five coins, 1 July to 31 December 2025 and 1 January to 30 June 2025; Nifty 50, 2024 and 2025	e-detector as E6 and E7; BOCPD with the robust variance of the previous day as prior mean and hazard $1/A$; rolling BJV-type test with $k = 120$; in a second run also BJV-type tests with $k = 480$ and 1440 and BOCPD with $w = 720$ and 1440 minutes (375 bars on Nifty); one-day pause after every alarm for every method; competitor thresholds rate-matched per series
E14	Binance five coins, July to December 2025; Nifty 50, 2024 and 2025; the E12 event design	as E6 and E7, bands 0 and 0.25; running and restarted monitors as in the synthetic runs
E17	Binance five coins, 1 July to 31 December 2025	as E6 with band 0.25 and recalibration after every alarm; detectors at 1, 5 and 15 minutes, three certified averages and the balanced maximum; settings committed before the first real run; rate-matched comparison (Q17g) added after an independent review

Appendix D. Correspondence with the accompanying repository

In this paper	In the repository
Lemma 1, Lemma 2, Theorems A and B, Proposition 1, Corollary C	`docs/math.md`, section 5.1 and section 5.4 (Lemma 1, Theorems A and B, Corollary C)
Proposition 2	`docs/math.md`, section 5.3; Corollary 3 (sequential form) is stated here
Theorem D.1, Theorem D, Corollaries E and F, Lemma A.1	`docs/math.md`, Theorem 6.2, Theorem 6.4, Corollaries 6.5 and 6.6, Lemma 6.1
Lemma 4, Lemma 5, Corollaries 6 and 7	`docs/theory_delay.md`, Lemma 1, Lemma 2, Corollaries 3 and 4
Theorem 8	`docs/theory_delay.md`, Theorem 1 and Lemma 5

In this paper	In the repository
Theorem 9	`docs/theory_delay.md`, Theorem 2(b) and Theorem 3; part (a) is stated here at the level of returns
Second-order formula (Section 4.3)	`docs/theory_delay.md`, section 4.3
Proposition 10	`docs/report_parts/E10.md`, Q10d
Section 5.5, Supplement S4.1	`docs/theory_running_tau.md`, `docs/report_parts/E14.md`
Section 5.6, Table 8	`docs/report_parts/E13.md`
Supplement S4.2	`docs/theory_panel.md`, `docs/report_parts/E15.md`
Section 5.5, Supplement S4.3	`docs/theory_multifreq.md` (Propositions 2 and 3, Theorems 1 and 3, Corollary 4), `docs/report_parts/E17.md`
Section 7	`docs/estimator_issues.md`, Issues 1, 2, 3 and 5
Code	`src/avmonitor/detectors.py` (bets and e-detectors), `monitor.py` (streaming monitor), `confseq.py` (confidence sequence), `forecast.py`, `design.py`, `running.py` (running truncation and restarts), `competitors.py` (BOCPD and the BJV-type test), `panel.py` (panel monitor), `multifreq.py` (detectors at several frequencies)
Experiments	`experiments/e04_detection.py`, `e04e_review_checks.py`, `e08_cs_baseline.py`, `e09_stress_guarantee.py`, `e10_forecast_relative.py`, `e11_delay_theory.py`, `e11f_split_c.py`, `e06a_five_coins_tables.py`, `e06b_five_coins_monitor.py`, `e07_nifty.py`, `e07c_india_vix.py`, `e07d_events.py`, `e12_crypto_events.py`, `e13_benchmarks.py`, `e14_running_tau.py`, `e15_panel.py`, `e17_multifrequency.py`; results in `results/`, report in `docs/experiment_report.md`
Figure 1	`paper/make_paper_figures.py`

Supplementary material

The supplement holds the detailed versions of results that the main text summarises. Section and table numbers refer to the main text unless they start with S.

S1. Second-order behaviour at practical alarm levels [HEURISTIC, checked numerically]

The first-order formula $\log A/g$ is a poor guide at $A = 7800$. For one bet and the change at the first step, $M_n = e^{S_n} V_n$ with $S_n = \sum_{i \le n} \log f_i$ and $V_n = \sum_{j=0}^{n-1} e^{-S_j} \uparrow V_\infty < \infty$, so $\log M_n$ is a random walk plus a slowly changing term. Nonlinear renewal theory (Woodroofe 1982; Pollak 1987 for the Shiryaev-Roberts statistic) suggests

$$\mathbb{E}[T_A] \approx \frac{\log A + \kappa - \zeta}{g}, \qquad \kappa = \frac{\mathbb{E}[H^2]}{2\,\mathbb{E}[H]}, \quad \zeta = \mathbb{E}[\log V_\infty],$$

with H the first ascending ladder height of S (so κ is the limiting mean overshoot). We have not checked the regularity conditions of those theorems for our law of Y (a continuous part and an atom at the cap), so we call the formula heuristic. Approximating S by a Brownian motion with drift g and variance s^2 per step, Dufresne's (1990) identity for exponential functionals gives $\zeta \approx \log(2/s^2) - \psi(2g/s^2)$, with ψ the digamma function. For a growth-optimal bet and a small change $s^2 \approx 2g$, so $\zeta \approx \log(1/g) + \gamma_E$ with Euler's constant $\gamma_E = 0.5772$, and

$$\mathbb{E}[\text{delay}] \approx \frac{\log(Ag) - \gamma_E + \kappa}{g}.$$

The memory term ζ is the reason the first-order formula fails: for a 25% increase in standard deviation $g \approx 0.05$, so $\zeta \approx 3.6$ while $\log A = 8.96$.

Table S1 compares the predictions with simulation (true baseline, $c = 3$, $A = 7800$, change at the first monitored step, 3000 paths per case, mean delays, E11) [SYNTHETIC].

Case	g_{grid}	first order $\log A/g_{\text{grid}}$	proved bound (Thm. 8a)	simulated, default grid	second order, best single bet	simulated, best single bet
Gaussian, sd ratio 1.1	0.0093	964	1221	455	405	417
Gaussian, sd ratio 1.25	0.0502	179	233	122	109	107
Gaussian, sd ratio 1.5	0.161	56	75	48	42	42
Gaussian, sd ratio 2	0.413	22	30	23	18.4	18.7
Jumps, sd ratio 1.25	0.0473	190	248	128	115	115
Student t_5, sd ratio 1.25	0.0320	280	362	178	158	159
Gaussian, sd ratio 0.8	0.0340	264	314	195	157	156
Gaussian, sd ratio 0.67	0.0598	150	179	124	99	99
Gaussian, sd ratio 0.5	0.0839	107	127	93	74	74

Table S1. Predicted and simulated mean delays in steps (E11, Q11a). Standard errors are below 1.5% of the mean.

In all 20 cases of E11 the second-order prediction for a single bet is within 3% of its simulated mean delay, the first-order formula overstates the delay of small changes by up to 112%, and the proved bound is 1.3 to 2.7 times the simulated delay. The default grid costs 9% to 26% against the best single bet. With the change after 500 pre-change steps, as in E4, the delays are 17% to 34% shorter, because the Shiryaev-Roberts statistic enters the change with a head start, and they match E4 to within about 10% (E11). The asymptotic regime arrives very slowly: at $A = 10^8$ even the best single bet is 5% to 24% faster than $\log A/\mathrm{KL}_{\text{inf}}$ in 15 cases (Gaussian, Student t_3 and jump returns), and the second-order formula explains the gap to within 0.024 in ratio for every $A \geq 10^4$ (E11, Q11e; Figure `e11_asymptotic_optimality.png`).

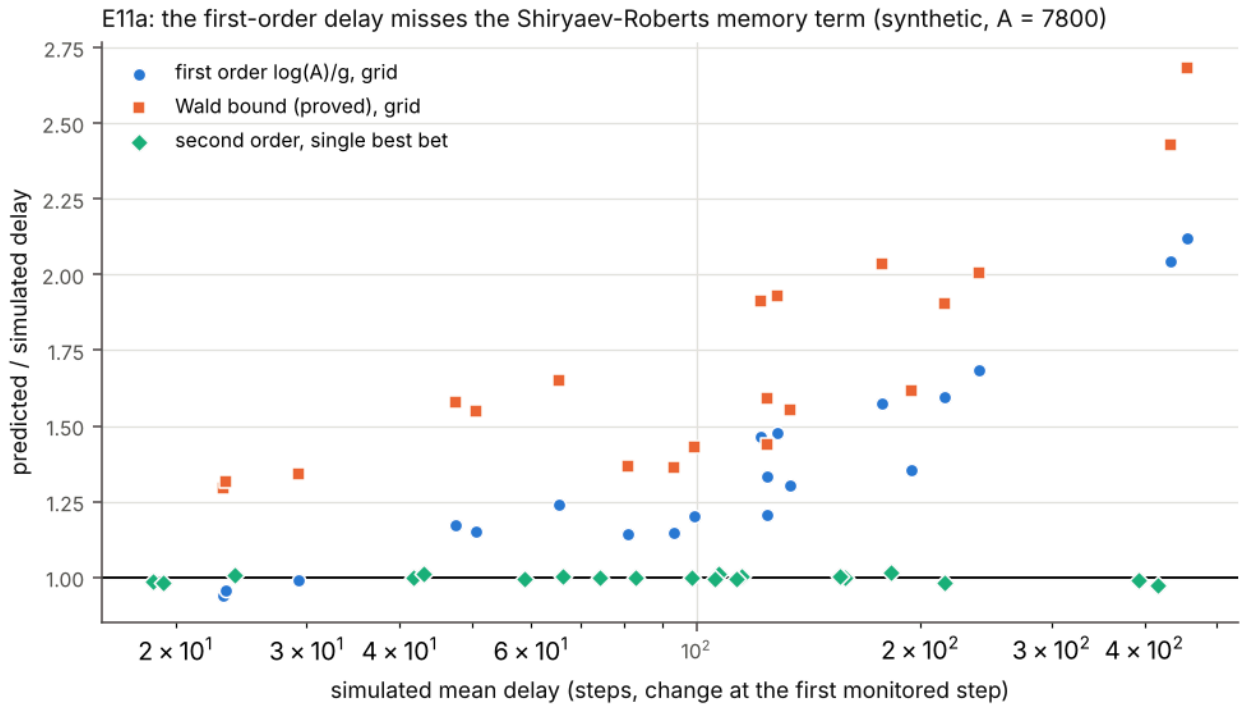


Figure S1. Ratio of predicted to simulated delay for the first-order formula, the proved bound of Theorem 8(a) and the second-order formula (E11) [SYNTHETIC].

S2. Delay details: truncation constant, slack of the bound and bet design

The truncation constant. For increases on clean data the delay barely changes for $c \geq 4$, and $c = 3$ costs 3%, 6% and 12% at sd ratios 1.25, 1.5 and 2 against no truncation. With jumps (probability 0.005 per step, standard deviation 8 times the diffusive one) the growth at sd ratio 1.25 peaks at c between 3.25 and 4.25 depending on the change size, and without truncation it falls to 29% of the $c = 3$ value. A burst of jumps at constant volatility (jump probability raised from 0.005 to 0.02) moves the $c = 3$ detector only 3.5% as much as a real 25% volatility increase, against 39% without truncation (E11). For decreases the speed limit dominates and a smaller c is usually faster: with the true baseline and Gaussian data, $c = 2$ shortens the mean decrease delay by 26% to 49% relative to $c = 3$ (E11). The two sides need not share c: with a known baseline, or with the confidence-sequence baseline, which the monitor supports for the split by running the lower bound on $\min(r^2, \tau^2_{\downarrow})$ and the upper bound on $\min(r^2, \tau^2)$, each at level $\delta/2$, each side is a valid e-detector for its own truncated statistic and the two-sided rule of Proposition 1 needs only each side's guarantee, so c of about 3 to 4 for increases and 1.5 to 2 for decreases is a legitimate design. The decrease null then concerns the mean of $\min(r^2, \tau^2_{\downarrow})$ for the smaller level $\tau_{\downarrow}$; volatility that has not fallen implies it under a fixed innovation shape (Lemma 1), but a smaller clip reacts more to changes of shape at constant volatility, such as a larger share of zero returns. In the experiments the split was run with a 1440-step plug-in baseline, which no proof covers: over all 18 no-change cells the one-day rate was 5.0% to 6.7%, and 5.0% to 6.05% with $c_{\downarrow} = c$, so the excess over the 5% bound that holds with the true baseline is there without the split and comes from the baseline (E11f). Table S2 gives the median decrease delays. Rounded prices use a tick fixed in units of the pre-change standard deviation, so that more returns are zero after a fall (at a tick of 0.5 standard deviations, 19.7% before the change and 24.5%, 29.1% and 38.3% after falls to

0.8, 0.67 and 0.5 of the standard deviation); a first version of the experiment let the tick shrink with volatility and overstated the gain.

Returns (calibration scale)	sd × 0.8: $c_{\downarrow}$ = 3 / 2 / 1.5	sd × 0.67	sd × 0.5
Gaussian	247 / 176 / 185	158 / 85 / 73	118 / 58 / 40
Student t_4	334 / 289 / 277	160 / 117 / 106	100 / 62 / 48
Jumps	273 / 196 / 206	161 / 88 / 74	119 / 59 / 40
Ticks of 0.25 sd (scale 1.11)	292 / 185 / 188	191 / 98 / 77	144 / 69 / 45
Ticks of 0.5 sd (scale 0.74)	**188 / 211 / 251**	99 / 80 / 86	70 / 39 / 35
Ticks of 0.75 sd (scale 1.11)	295 / 201 / 207	192 / 100 / 81	144 / 69 / 47

Table S2. Median decrease delay in steps with a separate decrease clip $c_{\downarrow}$ and $c = 3$ for increases, plug-in baseline from 1440 steps, $A = 28,800$, change at the first monitored step; 600 paths per cell for the continuous laws and 2000 for rounded prices, the same paths for the three values of $c_{\downarrow}$; every path detected. "Scale" is the median calibration scale (scaled MAD) in units of the true pre-change standard deviation (E11f) [SYNTHETIC].

With $c_{\downarrow} = 2$ the median delay fell in all nine cells of the continuous laws, by 13% to 51% (for Gaussian data the mean delay fell by 25%, 44% and 50%, close to the 26%, 43% and 49% that the theory predicts with the true baseline), and in all six cells with ticks of 0.25 and 0.75 standard deviations, by 32% to 52%. At a tick of 0.5 standard deviations it was 12% slower for a 20% fall (median 211 against 188 steps; paired difference of 29 steps, standard error 1.7; slower on 66% of paths) and faster for larger falls, by 19% and 44%. $c_{\downarrow} = 1.5$ was no better than $c_{\downarrow} = 2$ for a 20% fall except under Student t_4 returns, and at the tick of 0.5 standard deviations it was 34% slower than $c = 3$. A reading of the tick case, not a proof: with rounded returns the calibration scale is, on almost every path, a whole number of ticks times 1.4826, so at a tick of 0.5 standard deviations it is 0.74 standard deviations against 1.11 at ticks of 0.25 and 0.75; $c = 3$ then already clips at about 2.2 standard deviations, near the best clip for a 20% fall, and $c_{\downarrow} = 2$ at about 1.5, past it, while the growing share of zero returns after the fall adds to the effect (E11f). On the five cryptocurrencies of Section 6, which have few zero returns (1.0% to 4.3% of deseasonalised returns), the choice $c_{\downarrow} = 2$ was neutral to slightly positive for the confirmation rate of decrease alarms, within sampling noise (E11f) [REAL]. On data with a coarse tick, $c_{\downarrow} = 2$ should be compared with $c = 3$ at that tick before it is used.

The slack of the false alarm bound. E11 locates the slack with an exact optional stopping identity that writes the ratio of the false alarm rate to m/A as a product of a stopping factor, the share F_2 of expected wealth held on paths that crossed A, and an overshoot factor F_3 (E11, Q11d). For the default two-sided monitor (40,000 paths, rate 1.03%) the stopping factor is close to 1, $F_2 = 0.24$ and $F_3 = 0.80$: the slack is overshoot (a factor of 1.25) times wealth stranded on paths that never alarm (a factor of about 4). Of the stranded wealth, 60% is held by the decrease side and 31% by the three smallest increase bets, which cannot build wealth within a day. The split of the budget between the two sides costs almost nothing (both sides alarm on the same day in 0.00% of days). The bound itself is nearly sharp: for a single increase bet of 0.5, dynamic programming over the whole nonparametric null finds a law with a one-day false alarm rate of 4.73%, 95% of

the bound, against 3.52% for Gaussian data. For the default mixture the worst law found reaches 42% of the bound; a certified sharper bound for mixtures is open. Provable gains are possible by splitting the budget unevenly between the sides and by dropping bets that cannot use it.

Bet design. The quarticity estimator can enter the detector through the bet. A plug-in bettor with $\lambda_t = (\hat{\mu}_t - 1)/(\hat{v}_t + (\hat{\mu}_t - 1)^2)$, where $\hat{\mu}_t$ and $\hat{v}_t$ are kernel-weighted local mean and variance of Y over the last 60 steps (so that the local second moment is a truncated local quarticity), added with weight one half to the grid, was slower than the grid alone in 14 of 18 cells, by up to 9% (E4, Q4c). E11 explains this negative result: the quadratic approximation behind the bet makes it 1.4 to 4.4 times too small for increases, and it keeps only 49% to 94% of the attainable growth (E11, Q11a). A grid designed by minimising the worst-case ratio to the best single bet over a range of changes (sd ratio 1.2 to 2 up, 0.5 to 0.85 down) and four tail shapes, $\lambda \in \{0.16, 0.39\}$ up and $\lambda/\lambda_{\max} \in \{0.5, 0.85\}$ down, was at least as fast as the default in all 48 validation cases in its design range, by 0.4% to 10.5% (median 5%), but up to 17% slower for small increases (sd ratio 1.1) under fat tails. A grid tuned on Gaussian data alone was up to 18% faster on Gaussian or jump data and up to 55% slower under Student t_3 data (E11, Q11c). The gains from bet design are modest, and tuning to one tail shape is risky.

S3. Clustering, forecasts and price noise

Volatility clustering. Under GARCH(1,1) returns with constant unconditional variance, the conditional mean of X_t moves above and below m_0, so the nulls of Section 3 are false. With the true unconditional baseline and $A = 28,800$ for a 1440-step day, the share of days with an alarm is 78% at persistence $\alpha + \beta = 0.95$ (half-life 14 steps) and 100% at 0.99, 0.999 and 0.9999 (E9, Q9c; Figure 2, right). A band $b = 0.5$ absorbs short-lived clustering (3.6% at persistence 0.95), but no reasonable static band absorbs persistent clustering: even $b = 1$ on the increase side leaves 12% to 20% of days with an alarm at persistence 0.99 or more. For slow log-normal stochastic volatility, a band of 0.5 brought the daily rates to 1.7% (increase) and 0.0% (decrease) while a doubling or halving of volatility was still detected on 97% and 100% of paths (E4, E4c); the band is a design choice, not a guarantee. These alarms are not errors of the proof: persistent GARCH clusters are changes in conditional volatility, and the monitor reports them.

Forecast-relative monitoring. Monitoring $r_t/\hat{\sigma}_t$ with a GARCH(1,1) forecast fitted on a separate path and then frozen brings the one-day false alarm rate down to 3.0% to 6.2% at all three persistences (600 paths, plug-in baseline from one day), the level of the oracle with the true conditional volatility and the same plug-in baseline (4.7% to 5.3%). This is not a proved guarantee: the proof needs the true or a confidence-sequence baseline and a correct forecast model, and with the true baseline the one-day rate for i.i.d. returns is 1.1% to 1.4% (E9), so the plug-in roughly quadruples it. An EWMA forecast is only as valid as its match to the clustering (half-life 360 steps: 72% to 100% at band 0) (E10, Q10a). The price is the signal cap of Proposition 10. At persistence 0.99, a doubling of the long-run variance was detected within one day on 34% to 43% of paths and within five days on 93% to 94%, with median delays of 1523 to 1791 steps; without a break the same monitors alarmed within one day on 4% to 6% and within five days on 38% to 44% of paths, the latter far above the 25% that Theorem B guarantees with a known baseline (E10, Q10b and Q10b2). A fourfold increase of the long-run variance was detected within a day on 89% to 90% of paths (median 612 to 712 steps). For the oracle, whose

standardised returns are exactly i.i.d., the excess over five days is the plug-in effect of Section 3.4, not a property of the forecast.

Price-level microstructure noise. With i.i.d. noise on log prices, $\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}]$ depends on the last noise value, so squared returns are positively autocorrelated and the null fails without any change in volatility. At one-second sampling (23,400 steps per day, A set to give 5% per day) the one-day false alarm rate of the increase side on raw returns rose from 5.0% without noise to 8.5%, 18.0%, 31.0% and 41.5% at noise-to-signal ratios 0.5, 1, 2 and 4 (200 days per cell); overlapping pre-averaged returns alarmed on every day, because consecutive windows share most of their prices. Pre-averaging over non-overlapping blocks of prices, skipping the one return between blocks, gives each block fresh noise values, so the null holds for the block statistic whenever the spot variance and the noise law are unchanged, and Theorems A and B apply to it for every block length $k \geq 2$; the block monitors were below the bound or within Monte Carlo error of it in all 58 cells tested, and with $k = 10$ were about 8 times slower than raw returns without noise and faster than them from a noise ratio of about 2 (ALT1) [SYNTHETIC]. One-minute crypto returns were not tested for noise of this kind; the effect is largest at the highest frequencies.

S4. Extensions (E14, E15, E17)

S4.1 A running truncation level and restarts with a proof (E14)

The monitor of Section 3 fixes τ before monitoring and recalibrates for a day after every alarm. E14 studies three constructions that relax this, with proofs in the accompanying theory note on running truncation levels. (A) A band $L(s) \leq m(s) \leq U(s)$ for the truncated mean $m(s) = \mathbb{E}[\min(r^2, s^2)]$ that holds for all truncation levels s at once with probability $1 - \delta$; the increase side then bets against $U(\tau_t)$ and the decrease side against $L(\tau_t)$ for any predictable τ_t, and $\mathbb{P}(\text{false alarm within } m) \leq \delta + m/A$. The band is built from empirical Bernstein confidence sequences on a grid of 13 levels, or from a time-uniform bound on the distribution function. (B) Smoothed conformal p-values, the ranks of $|r_t|$ among all earlier absolute returns, which are i.i.d. uniform when the returns are exchangeable (Vovk, Nouretdinov and Gammerman 2003); betting on them gives $\mathbb{P}(\text{false alarm within } m) \leq m/A$ exactly, with no baseline, no δ and no τ. Its null, exchangeability, is stronger than the conditional-mean null, and it reacts to any change in the law of $|r_t|$, not only to a change of scale. (C) Restarts with no blind period, which bet from the first return after an alarm. The band version learns its band from the returns after the restart while monitoring; the Shiryaev-Roberts chain started at step j bets against the band as it stood before j, multiplied by the indicator that the band has not failed by then, an event known when the chain starts, which keeps each chain a supermartingale. The rank version restarts its reference set empty and bets from the first return after the alarm. Its bound m/A then holds for the new segment if the returns of the segment are exchangeable conditionally on the returns up to the alarm; the condition is stated given the past because the alarm time depends on the data. It holds for i.i.d. returns and for any infinite exchangeable sequence (Remark B.3 of the theory note).

The results [SYNTHETIC] (5000 null paths per cell, the four laws of E9, every proved bound 5% per 390-step day) sharpen Section 3.4. Within one day the plug-in stayed below 5% (4.0% to 4.9% with 390 calibration steps), but within five days it alarmed on 40.6% to 42.9% of null paths, where a known baseline would guarantee at most 25%, with or

without a running τ. Section 3.4 states the explanation of the theory note: at a fixed calibration length the plug-in has a positive drift under the null on a share of calibration windows that does not depend on A, so that its run length there grows only like $\log A$; Lemma A.7 proves that this share is positive, and its size (77% to 83% at 390 calibration steps, 56% to 68% at 1440) is a numerical estimate (Q14w). The rank detectors, with likelihood-ratio bets in rank space or with indicator bets, were valid at their nominal level and used their budget (1.7% to 3.2% per day, 14% to 18% over five days, against 0.8% to 1.7% and 12% to 14% for the oracle with the true baseline), while the band of route A and the confidence sequence of Section 3.4 were valid with a very wide margin (at most 0.26% over five days). In delay, the rank detector with likelihood-ratio bets beat the plug-in on every decrease, by a factor of two to three for the larger ones (sd ratio 0.5: 25 to 34 steps against 69 to 77), partly because its bets are clipped likelihood ratios for a normal scale change, while the bets on X are linear with λ at most 0.5; it was slower on increases under Student t_3 returns (sd ratio 1.25: 243 to 260 steps against 182 to 194). Route A with 1440 calibration steps was close to the confidence sequence on increases and faster on decreases with a running τ (Gaussian, sd ratio 0.5: 71 steps against 105), but with 390 calibration steps it missed most small changes. On small increases the running τ costs a little for Gaussian returns (1440 calibration steps, sd ratio 1.25: 336 against 326 steps with τ fixed) and more under Student t_3 tails (937 against 783 steps, and 76% against 88% of changes detected), because after an increase τ_t moves up to a level where the band is wider (E14, Q14a and Q14b).

Restarts without a blind period [SYNTHETIC]. After a restart with no change (4000 paths per cell, the four laws of E9), the rank version alarmed within 390 steps on 2.70% to 3.35% of paths and within 1950 steps on 15.7% to 17.0% (bounds 5% and 25%), and the band version on at most 0.03% (bounds 5% and 21%). With a change 120 steps after the restart, the rank version detected it in a median of 20 to 70 steps for Gaussian returns and 42 to 115 steps for Student t_3 returns, on 65% to 99% of paths; the standard protocol, which calibrates on the first 1440 returns of the segment, found at most 61% of these changes within 2000 steps, 1660 to 1830 steps after the change. Even a change 30 steps after the restart was often found (halving of the standard deviation: 91% of Gaussian and 70% of Student t_3 paths). The band version detected nothing when the change came 30 steps after the restart, because its band from so few returns is very wide and later snapshots already contain post-change returns, and at most 34% of changes other than a halving when it came after 120 steps; where both detected, the rank version was about 3 to 10 times faster (E14, Q14f).

Real data [REAL]. On the five coins (July to December 2025, calibration 1440 minutes, band 0.25) and the Nifty 50 (2024 and 2025), the standard protocol monitored only 30% to 33% of crypto minutes (40% for the Nifty 50), while the restarted monitors watched all of them and raised 0.66 to 0.88 alarms per crypto day (0.23 to 0.25 per Nifty session), about as many as the standard protocol (0.66 to 0.70). The band 0.25 has two meanings here. For the monitors that bet on X, the band version included, it widens the baseline by 25% in each direction; for the rank version it divides each payoff by 1.25, which tolerates a conditional mean payoff of up to 1.25 and has no direct meaning in units of volatility. E12's setting, 0.25, is used for both (E14, Q14c). Pooled over the six series, 80% (band version) and 74% (rank version) of their increase alarms and 85% and 90% of their decrease alarms were followed by a next day on the flagged side of the baseline, against 82% and 90% for the standard protocol. Without a band the rank version alarmed 3.6 to 4.3 times per crypto day with lower confirmation rates (67% and 73%): right after a

restart its reference set is small, and it soon rejects exchangeability, which clustered volatility violates. Among the monitors that recalibrate, the proved ones alarmed less than the plug-in and were confirmed about as often after increase alarms (78% to 83% against 82%) and more often after decrease alarms (94% to 97% against 90%) (E14, Q14d).

Month-long streams [SYNTHETIC]. In 40 synthetic streams of 30 days of 1440 steps per cell, with every monitor restarting after each alarm at the thresholds of Section 6 (band 0): with no change, the plug-in raised 0.094 (Gaussian) and 0.084 (Student t_3) false alarms per day against a nominal 0.05, while the proved monitors stayed within their per-day bounds (rank restart 0.029 and 0.035 against 0.050; band restart and confidence sequence essentially 0). With the standard deviation multiplied by 2, 1.5, 0.67 or 0.5 about once a day, the monitors that recalibrate for a day after each alarm were blind 57% to 71% of the time, awake at 31% to 49% of the changes, and detected 56% to 66% of them within a day (65% to 66% for the default plug-in; median delay 78 to 166 steps); the rank restart detected 99.6% and 98.0% (median 34 and 60 steps) and the band restart 94.3% and 77.8% (median 97 and 153) (E14, Q14h).

Event study rerun [REAL]. With the design of Section 6.3 unchanged except for the monitor, the restarted monitors were watching at all 114 releases (48 FOMC statements and 66 CPI releases over three coins), against 55 for the streaming plug-in. Up alarms within six hours rose from 26 of 48 FOMC statements (plug-in) to 30 (band restart) and 35 (rank restart), and from 21 of 66 CPI releases to 24 and 31; every awake cell had a stratified $p \le 0.006$ against matched controls, including BTC CPI ($p = 0.003$ and 0.0003, against 0.11 for the plug-in). The rank restart paid with a higher control rate (13% to 19% of matched control windows, against 7% to 14% for the band restart and 8% to 17% for the plug-in), so part of its gain is more alarms everywhere. Being awake did not mean flagging every release, and compared at the same releases the restarted monitors flagged fewer. At the 26 FOMC statements at which the plug-in was monitoring, it flagged all 26, the band restart 19 and the rank restart 23; at the 29 CPI releases at which it was monitoring, it flagged 19 (its 21 above include two releases at which it woke up within the six hours), the band restart 10 and the rank restart 14. A monitor that restarted shortly before a release compares the release against a short or volatile baseline, which may explain part of the gap; we did not test this (E14, Q14g).

Cost [SYNTHETIC]. On a shared machine, in plain Python with NumPy, one return at a time: 13 microseconds per return for the plug-in, 15 to 16 for the running band, 44 for the rank monitor (whose sorted reference set grows when there is no alarm) and 47 for the band restart (77 in its learning phase); only the ratios are informative (E14, Q14e). On the real series above, measured later while the machine was more heavily loaded, the band restart cost 97 to 182 microseconds per return, because it restarts its learning phase after every alarm, the rank restart 45 to 70 and the monitors that recalibrate 5.8 to 45.6 (E14, Q14d). Every monitor costs well under a millisecond per return.

S4.2 Many assets: false discovery control (E15)

A user rarely watches one asset. With K monitors at level α per day each, the chance of some false alarm on a quiet day grows to about $K\alpha$, and Bonferroni's correction slows every monitor. The e-BH procedure (Wang and Ramdas 2022) controls the false discovery rate for e-values under any dependence, but a Shiryaev-Roberts statistic is not an e-value at a data-dependent time. E15 therefore gives each asset a mixture e-process for "no

change since the block started", $E_n = \sum_{j \le n} w_j L_n^{(j)} + \sum_{j > n} w_j$ averaged over bets and sides (weights w_j over start times, fixed in advance, summing to 1), applies e-BH to these e-processes at every step, and stops each asset at its discovery ("sticky e-BH"). Within one block and whatever the dependence between assets, the false discovery rate is at most α at every stopping time, the expected largest false discovery proportion over the block is at most $\alpha(1 + \log(1/\alpha))$, about 20% for $\alpha = 5\%$, and with no change anywhere the probability of any discovery is at most α (accompanying theory note on panel monitoring, Theorem 1). The proof needs each asset's null to hold given the past of the whole panel, which is stronger than its own-past null when volatility spills over between assets with a lag.

In simulation [SYNTHETIC] (true baselines, blocks of 390 steps, $\alpha = 5\%$, independent assets or a common factor with pairwise correlation 0.7 and common jumps), the naive rule raised a false alarm in 37.2% of quiet blocks with 50 independent assets, while e-BH and Bonferroni stayed at 0.3% to 0.9% for K from 5 to 50. With changes, the false discovery rate of e-BH stayed below 1% in all 48 designs, and e-BH found on average 5.0 percentage points more changed assets than Bonferroni (at most 21.3 points more), but its probability of at least one false discovery, which it does not control, reached 8.0% at $K = 50$. Two negative results: with the confidence-sequence baseline the power for a 25% increase fell from about 93% to 36% to 41%, much more than the multiplicity correction costs; and a guarantee per daily block is not a guarantee for a horizon of ten blocks, over which daily e-BH had a 23.5% chance of at least one false discovery against 4.25% for daily Bonferroni (E15).

S4.3 Several sampling frequencies (E17)

Set-up. Frequency $k \in \{1, 5, 15\}$ (minutes) sums the one-minute returns of non-overlapping k-minute blocks or, in the *gap* version, the last $k - 1$ returns of each block, so that consecutive blocks share no price. Each block return $R_b^{(k)}$ gives $X_b^{(k)} = \min((R_b^{(k)})^2, \tau_k^2)$, with τ_k three scaled MADs of k-minute returns, and frequency k runs its own increase and decrease detectors over blocks with the default grids; between block ends its statistic $M^{(k)}$ keeps its last value. The weights $w_k \ge 0$ sum to 1. Each frequency needs its own null, $\mathbb{E}[X_b^{(k)} \mid \mathcal{F}_s] \le m_k^{\uparrow}$ for every block (and $\ge m_k^{\downarrow}$ on the decrease side), where s is the minute before the block starts and $\mathcal{F}_s$ holds all one-minute information up to s. The one-minute null does not imply it in general: if every return of a block takes a fair random sign drawn for that block, $r_t = S_b|e_t|$ with e_t i.i.d. standard normal, the one-minute null holds exactly but a 5-minute block has mean square 17.7 instead of 5. It does hold with the true baselines for i.i.d. returns with any tails or jumps, and for gap blocks under i.i.d. price noise on an efficient price with independent increments. The statements below are proved in the accompanying theory note on multi-frequency monitoring; we keep its numbering.

Why the usual proof fails (Proposition 2 of the note). A k-minute detector is an e-detector on its own clock, not on the one-minute clock: with $k = 2$, i.i.d. Gaussian returns, no truncation and one increase bet λ, a stopping time on the one-minute clock gives $\mathbb{E}[M_\tau] = 1 + p + 0.484\lambda$, while the expected number of completed blocks is only $1 + p$, with $p = \mathbb{P}(|Z| > 1) = 0.317$. This breaks the proof of Theorems A and B for the plain average checked every minute; it does not show that the average exceeds the bound.

Certified average (Theorem 1 and Corollary 4 of the note). Between block ends, bet j of frequency k is credited with $C_t = \min(M_s, \ell_j(M_s + 1))$, where M_s is its value at the last

block end and ℓ_j its smallest possible factor ($1-\lambda_j$ for an increase bet, $1-\varphi_j$ for a decrease bet); at block ends $C_t = M_t$. Each side averages C_t over its grid and over frequencies with weights w_k, and the monitor alarms when either side reaches $2A$. If every frequency with positive weight satisfies both nulls with the baselines that the monitor uses, then for every $m \geq 1$

$$\mathbb{P}(\text{alarm within } m \text{ minutes}) \leq \frac{1}{A}\sum_k w_k \lceil m/k \rceil,$$

and the mean time to the first alarm is at least $(A-\kappa)/\rho$ minutes, with $\rho = \sum_k w_k/k$ and $\kappa = \sum_k w_k(1-1/k)$. With recalibration after every alarm, the expected number of alarms within m monitored minutes is at most $(m\rho+\kappa)/(A-\kappa)$. *Sketch.* $\mathbb{E}[M_{e(t)} \mid \mathcal{F}_t]$, with $e(t)$ the end of the current block, is at least C_t and grows in mean by at most 1 per block started, so Ville's inequality applies.

Plain average and weighted maximum (Theorem 3 of the note). With frozen values checked every minute and $N_k = \lfloor m/k \rfloor$, a union bound gives $\sum_k w_k N_k/A$ for the rule $\max_k w_k M^{(k)} \geq 2A$ and $(\sum_k \sqrt{w_k N_k})^2/A$ for the plain average, three times the certified bound when $w_k \propto k$ ("balanced" weights). Neither has a restart result.

The price of not knowing the frequency (Proposition 3 of the note). If only frequency j sees a change, the certified average alarms when $M^{(j)}$ reaches $h_j = \sum_k w_k \lceil m/k \rceil/(w_j \lceil m/j \rceil)$ times the threshold that j would need alone at the same budget. For any weights $\max_j h_j \geq 3$, the cost of a Bonferroni split, with equality for balanced weights; equal weights give 1.27, 6.33 and 19. For 5% per 1440-minute day, A is 28,800, 5760 and 1920 for one frequency alone, 12,160 for equal and 4114 for balanced weights (E17, Q17a).

Table S3. Gaussian returns, true baseline, levels for a proved 5% per 1440-minute day (15% for the plain average); 10,000 null days, 2000 paths per change (E17, Q17a to Q17d) [SYNTHETIC]. Persistent changes: median delay in minutes (share detected if below 99.5%). Burst (standard deviation tripled for 15 minutes) and autocorrelation jump (lag-one autocorrelation from 0 to 0.3, one-minute variance unchanged): share detected (median delay).

Monitor	A	False alarms per day	sd ×1.25	sd ×0.5	Burst ×3, 15 min	Autocorrelation jump
1 min	28,800	1.96%	134	93	73.2% (12)	3.8% (707)
5 min	5760	0.71%	508 (98%)	403	0.8% (18)	99.2% (448)
15 min	1920	0.39%	893 (57%)	1103	0.0%	81.6% (758)
Average, equal weights	12,160	1.50%	138	96	70.3% (12)	95.5% (593)
Average, balanced	4114	0.77%	153	107	62.7% (13)	97.3% (525.5)
Average, $w_k \propto \sqrt{k}$	6909	1.08%	143	100	68.2% (12)	96.6% (550.5)
Plain average every minute, balanced	4114	0.78%	153	107	62.7% (13)	97.3% (523)

Monitor	A	False alarms per day	sd ×1.25	sd ×0.5	Burst ×3, 15 min	Autocorrelation jump
Weighted maximum, balanced	4114	0.72%	153	107	62.6% (13)	97.0% (533)
Average of 5 and 15 min, balanced	2880	0.50%	563 (98%)	448	0.4% (18)	98.5% (493)

Results [SYNTHETIC]. With the true baselines and i.i.d. Gaussian, Student t_3 or jump returns, every monitor alarmed on 0.32% to 2.62% of days, and the plain every-minute average stayed within 0.07 points of the certified one: the factor 3 is a gap in the proof, not in behaviour. With plug-in baselines the one-minute detector alarmed on 5.47% of Gaussian days, significantly above 5%, and the balanced average on 3.66% to 4.54% (Q17b). Under price noise ($r_t = e_t + u_t - u_{t-1}$, $u_t \sim N(0, \gamma^2)$ in units of the one-minute efficient standard deviation, as in ALT1) with true baselines, the one-minute detector alarmed on 6.88% and 12.06% of days at $\gamma = 1$ and 2, the balanced average on 3.09% and 6.50%, and the proved gap average of 5 and 15 minutes on 0.54% and 0.52%. At $\gamma = 2$ it detected persistent increases of 25% and 50% more often than either proved single gap detector (28.1% against 21.4% and 25.3%; 93.3% against 85.5% and 82.8%) but halvings less often than the 5-minute gap detector (47.2% against 56.5%); at the planned $\gamma = 1$ it did not help for increases (78.7% against 79.2% at sd ×1.25), and at neither level for decreases or a tripling lasting 60 minutes. The level $\gamma = 2$ was added after the $\gamma = 1$ results had been seen (Q17e).

Results [REAL]. On the five coins (July to December 2025, deseasonalised as in E6, band 0.25, plug-in baselines, recalibration after every alarm), the one-minute detector made up 98.4% to 99.7% of the combined statistic at alarms on average. The averages alarmed 0.65 to 0.68 times per day (one-minute monitor 0.68), and their up and down alarms were followed by next-day volatility on the flagged side of the baseline 78% to 80% and 84% to 88% of the time, against 80% and 88%. The 5- and 15-minute detectors alone alarmed 0.43 and 0.26 times per day and were confirmed more often (up 90% and 95%, down 95% and 96%) (Q17f). In a comparison added after an independent review, the one-minute monitor with its two levels set per coin to give about as many up and down alarms as the 5-minute detector had its up alarms confirmed 90.5% of the time against 90.0% ($p = 0.88$) and its down alarms 96.4% against 94.6% ($p = 0.43$); matched to the 15-minute detector, its down alarms were followed by volatility below an EWMA forecast 88.9% of the time against 72.4% ($p = 0.0014$). The p-values treat alarms as independent and are optimistic (Q17g).

S5. Real-data details

S5.1 The Nifty 50 index and India VIX

Data. Nifty 50 one-minute bars from April 2017 to October 2026, sessions of 375 bars, overnight returns excluded. Stretches of five or more minutes with a frozen index value (exchange outages, the two market-wide halts of March 2020, and the closing auction minutes from 3 August 2026) were set to missing, leaving 2,361 sessions and 882,001 returns. The intraday profile is session-aware, so that the close is not smoothed into the open. Pre-registered configuration: calibration 750 bars, band 0.25; development period

2017 to 2021, held-out period 2022 to October 2026, with nothing changed after the held-out period was seen (E7).

Results. The monitor raised 0.26 alarms per session in both periods. In the held-out period, 78% of increase alarms were followed by a session more volatile than the baseline (47% at random times) and 76% of decrease alarms by a calmer one (53%); the Gaussian likelihood-ratio monitor's increase alarms were followed by higher volatility only 61% of the time. India VIX agreed with the alarms on the same day (in the development years VIX closed higher on 68% of days with an increase alarm, against 47% of all days) but did not move in the alarm's direction on the next day more often than on other days (45% to 48% against 46% to 47%): the agreement is concurrent, not a lead. Among event days, 9 of 11 Union Budget sessions had an alarm, against 25% of ordinary sessions (Fisher $p = 0.0001$), and all 7 budget sessions at which the monitor was already watching at the open; Reserve Bank of India policy days were not different (32% against 25%, $p = 0.36$), consistent with their volatility being close to the recent norm (median 1.06 times the previous 20 sessions). Monthly derivative expiry days had more alarms (38%, $p = 0.002$), which may be an artefact of a profile that pools all weekdays. In March 2020 volatility was 2.7 times that of the previous 20 sessions, yet only 6 of 21 sessions had an alarm: after each alarm the monitor recalibrates on the new, already high level. It answers "has volatility changed from the recent baseline?", not "is volatility high?".

S5.2 FOMC statements and CPI releases

Design. Binance BTC, ETH and SOL from 21 January 2024 to 31 December 2025; the 16 FOMC rate decisions (statement at 14:00 US Eastern) and 22 US CPI releases (08:30 US Eastern) of 2024 and 2025 inside the monitored period, with every date checked against a published schedule. Control windows are all days at least two days away from any event day, with the same weekday, at the same Eastern clock time (134 for FOMC, 299 for CPI); each event is compared with the control windows of its own weekday, and the control rates below are these matched rates averaged over events. The outcome is an increase alarm within six hours of the anchor, and the test is an exact stratified permutation test. The primary family, fixed before the first run, is FOMC and CPI on BTC in the streaming design (the monitor as a user would have run it), with a Holm correction; ETH and SOL are replications, and a design in which a fresh monitor is calibrated on the day before every anchor was added after the streaming results were seen and is secondary (E12).

Results. In the streaming design, 9 of 16 FOMC statements on BTC were followed by an increase alarm within six hours, against 14% of control windows (Holm-adjusted $p = 0.0002$); the CPI test was not significant on BTC (5 of 22 against 12%, $p = 0.11$) but replicated on ETH ($p = 0.003$) and SOL ($p = 0.00004$). The streaming monitor was recalibrating after its own alarms about 70% of the time and was awake at only 9 of the 16 FOMC statements on BTC; at those it alarmed every time (and at 8 of 8 on ETH and 9 of 9 on SOL). With a fresh calibration, 14 to 15 of the 16 FOMC statements raised an alarm within six hours, against 29% to 32% of controls, and 59% to 73% of CPI releases, against 23% to 28%. The alarms came after the release (median 23 to 34 minutes, never before 14 minutes on BTC and SOL and 10 minutes on ETH), and there is no sign of more alarms in the hours before a release. The Gaussian likelihood-ratio monitor with a fresh calibration also alarmed at almost every release, but with a median delay of 0 to 2 minutes, on a single return of 8 to 9 robust scale units (median): one large minute is enough to trigger it. It also alarmed in 62% to 76% of control windows. Because it ran at band 0 and the robust

monitor at band 0.25, this contrast mixes the band with the truncation; run at band 0 as well, the robust monitor with a fresh calibration alarmed at 14 to 15 of 16 FOMC statements and 16 to 18 of 22 CPI releases, with a median delay of 18 to 22 minutes, and in 39% to 51% of control windows. The truncated detector cannot be moved by one minute (one clipped observation multiplies its statistic by a bounded factor), and its delays and the size of its largest recent return are those of a short volatility regime; the share of release alarms classed as “jump only” did not separate the two monitors, so this reading rests on the delays and the return sizes.

S5.3 A panel of assets

The panel monitor of Supplement S4.2 was run on the five coins with one block per UTC day, $\alpha = 5\%$ per day, the plug-in baseline of the previous 1440 returns and band 0.25 or 0.5, and on the five coins together with the Nifty 50 index, each asset on its own clock (E15) [REAL]. Three findings stand out. First, the strict daily null is false on most coin-days: at band 0.25 e-BH discovered 86% to 87% of coin-days, so the false discovery guarantee is nearly empty in practice, for the reason discussed in Section 6.5. Second, the multiplicity rule changed the timing more than the set: e-BH found only 3.7% to 8.7% more coin-days than Bonferroni, but on the coin-days found by both it flagged first 71% to 77% of the time (median 2 to 4 minutes earlier). Third, discoveries pointed the right way: an increase discovery was followed by a higher robust volatility over the next day 78% to 83% of the time against 46% to 50% at matched random times, and a decrease discovery by a lower one 85% to 97% of the time against 49% to 55%. As in Section 6.1, a per-asset threshold chosen with hindsight to make as many discoveries as e-BH had them confirmed at the same rate (no significant difference in any of 12 cells); the e-BH level is fixed in advance and carries a guarantee when the baselines are known. With the Nifty 50 in the panel, its discoveries were confirmed 87% to 95% of the time, and four times (band 0.25) e-BH discovered it after its session had closed, because crypto discoveries later in the day lowered the e-BH bar below its frozen e-value.